Big Data Engineering — A.Y. 2025/2026
Social Network Analysis Challenge, Track 2

PolySignal

A Social–Intelligence Layer for Polymarket Prediction Markets

Social Media Signals and Prediction Market Outcomes: a data-driven study on Polymarket

technical report

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Research question. *Do social-media signals provide incremental predictive value over market-implied probabilities, across thematic domains, platforms, and time windows? We answer it with a reproducible, leakage-controlled pipeline over 300 resolved Polymarket contracts and a corpus of ∼1.36 million social-media posts.*

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Abstract

Prediction markets such as Polymarket aggregate the beliefs of many traders into a single, continuously updated probability of a future event. Social-media platforms, in parallel, host the public discourse through which those beliefs form. This report studies, empirically and conservatively, whether that discourse carries information that is *not already priced in* by the market.

We build an end-to-end Big Data pipeline that (i) collects **300 resolved binary Polymarket contracts**, balanced across politics, finance, and sports (100 each), together with their **full-lifetime price series** (1,353,114 price points); (ii) collects a longitudinal social corpus of **1,368,041 posts** through source-appropriate collectors for the **three platforms** named by the assignment — Telegram (MTProto), Reddit (public-web), and X/Twitter (authenticated automation), **all three present in the analysed corpus**. Telegram dominates (98.6%), Reddit is secondary (1.2%), and X/Twitter is a small but real third platform (2,822 posts, 0.2%) whose scale is capped by X's anti-automation defences — a limitation we document rather than paper over (Section 1.4); (iii) links posts to contracts with a deterministic semantic-relevance scorer; (iv) enriches the text with sentiment, named entities, hashtags, language detection, and topic models; (v) projects the result into a Neo4j knowledge graph; and (vi) derives a per-market *Social Pressure Index* that feeds a leakage-controlled backtest.

The descriptive analysis confirms textbook market behaviour: market-implied probabilities are already strongly informative of the final outcome, and become monotonically more so as resolution approaches (AUC rising from 0.90 at 90 days to 0.99 at close). The predictive study delivers a clear and honest **null result**: a market-only model attains a Brier score of 0.031 and AUC 0.989, whereas adding social features *degrades* performance (ΔBrier = +0.009), and a social-only model is no better than chance (AUC 0.404). We trace this to three structural causes — a hype/optimism bias in social discourse, a coverage mismatch between the available channels and the market sample, and the informational efficiency of the market price itself — and we argue that this negative result is a meaningful empirical contribution rather than a failure. All findings are exposed through an interactive **React** dashboard served from a local, zero-dependency API.

Reproducibility note. Every quantitative result in this report is regenerated from the persisted pipeline artifacts by a single released step (`python scripts/_paper_figs.py`), so the figures and tables always match the dataset on disk and no number is hand-transcribed.

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CHAPTER 1

Introduction

1.1 Context and motivation

Prediction markets are exchanges on which participants trade contracts whose payoff is tied to the outcome of a future, verifiable event. Because a rational trader buys a "YES" share whenever its price is below their subjective probability of the event and sells when it is above, the equilibrium price of a binary contract behaves as a continuously updated, money-weighted estimate of the event's probability [2, 20]. Among contemporary platforms, *Polymarket* has become one of the most liquid and widely watched venues, hosting contracts on politics, macroeconomics, crypto-asset price levels, sports, and current affairs. The resolved history of these contracts provides something rare in forecasting research: a large set of events with a *known ground-truth outcome* and a *full price trajectory* leading up to it.

In parallel, a large share of the public reasoning *about* these same events happens on social-media platforms — Reddit threads, Telegram channels, and posts on X (formerly Twitter). This discourse encodes sentiment, narratives, rumours, and occasionally genuine private information. A long line of work has asked whether such signals anticipate real-world or market outcomes, with results ranging from optimistic [4, 16] to sharply qualified [19]. The question is far from settled, and it is exactly the question this project investigates in the specific, well-instrumented setting of Polymarket.

1.2 Research question and objectives

The project is organised around a single, deliberately conservative research question:

> **Do social-media signals provide *incremental* predictive value over market-implied probabilities, and if so, in which thematic domains, on which platforms, and over which time windows?**

The emphasis on *incremental* value is the methodological core of the work. It is easy — and uninteresting — to show that social sentiment correlates with an outcome when the market price already encodes that information. The scientifically meaningful test is whether social features improve a forecast *on top of* the market price, under an evaluation that forbids any leakage of post-resolution information. We therefore frame PolySignal not as an attempt to "beat Polymarket", but as an analytical layer that quantifies social context and submits it to a controlled, falsifiable test.

Operationally, the work pursues the three goals set out by the assignment:

1. **Dataset construction (Task 1).** Build a reproducible ingestion pipeline that collects resolved Polymarket contracts and their price time series, collects longitudinal social discourse, links posts to contracts, normalises and enriches the text, and stores everything in a queryable format.

2. **Analytical study (Task 2).** Characterise the contracts and the discourse descriptively; study the temporal relationship between social signals and price movements; compare platforms and domains; and expose the results through an interactive dashboard.
3. **Outcome prediction (Task 3, optional).** Engineer social features and evaluate, with temporal cross-validation and explicit ablations, whether they add predictive value over the market price.

1.3 Design philosophy

Four principles shaped every component and are worth stating up front, because they explain many concrete choices later in the report.

Deterministic core, optional intelligence.
The full pipeline runs end-to-end *without* any call to an external language model. LLM-assisted steps exist only as opt-in, cached verifiers layered on top of a deterministic baseline. This guarantees reproducibility and decouples our results from a paid, non-deterministic dependency.

Time discipline and no leakage.
Every record carries a timezone-aware timestamp, and every forecasting or backtesting step enforces a strict "no information from the future" rule: a market's resolution can never reach a feature, and a post published after a contract closed can never count as evidence for it.

No invented data.
The system never fabricates posts, markets, or labels to inflate coverage. Empty or sparse cells are reported honestly — a property that proves essential when we discuss the coverage limitations in Chapter 15.

Honest reporting.
We report a negative predictive result in full, with its diagnostics, rather than selecting a favourable slice. A robust null result, in a field prone to over-optimistic claims, is itself a contribution.

1.4 Scope decisions

Two scope decisions are load-bearing and are justified in detail where they arise; we summarise them here so the reader can keep them in mind.

Platforms: all three collected, Telegram-dominated. The assignment names three platforms, and the pipeline implements a collector for each: **Telegram** (via the official MTProto API, using Telethon), **Reddit** (via compliant public-web scraping with Scrapling against `old.reddit.com`, since we deliberately avoid the OAuth API), and **X/Twitter** (via authenticated browser automation with Playwright, because X's free programmatic tier no longer supports research collection). Each post, regardless of source, is normalised into a single platform-agnostic `SocialPostRecord`, so linking, enrichment, and evaluation never branch on

platform. **All three platforms are present in the analysed corpus**, but in sharply unequal volume: Telegram dominates (98.6%), Reddit is secondary (1.2%), and X/Twitter is a small third platform (2,822 posts, 0.2%). X's anti-automation defences (token challenges, login walls, aggressive rate limiting) cap how much can be gathered responsibly, so its contribution is modest — but it is genuine, and the cross-platform analysis of Chapter 11 is a real **three-platform** comparison (114 contracts carry all three sources). The load-bearing caveat is therefore not that a platform is missing but that the corpus is **Telegram-dominated and crypto/macro/geopolitics-tilted**; we restate this wherever it bears on a result rather than softening it.

Contracts: binary Yes/No only. We restrict the study to binary YES/NO contracts. Every downstream component — stance detection, the Social Pressure Index, the forecast blender, and the backtest — is natively two-outcome aware, and generalising to multi-outcome markets would require changing all of them in lockstep. Binary contracts are also the cleanest setting for the incremental-value question, since the target is a single Bernoulli variable.

1.5 Contributions

This report contributes:

1. a **reproducible Big Data pipeline** that turns two heterogeneous, high-volume sources (a prediction market and multi-platform social discourse) into a single, queryable, leakage-aware dataset of 300 resolved contracts and ∼1.37 million posts;
2. a **deterministic contract–post linking and outcome-aware stance** method that scales to hundreds of thousands of candidate pairs without an LLM in the loop;
3. a per-market **Social Pressure Index** that condenses volume, stance, engagement, source diversity, and temporal dynamics into interpretable features;
4. a **leakage-controlled backtest** with four baselines and an ablation that isolates the incremental value of social signals; and
5. a careful, evidence-backed **negative result** together with a root-cause analysis explaining *why* public social signals do not beat the market price in this setting.

1.6 Structure of the report

The report is organised into four parts that follow the data as it flows through the system. **Part I** (this part) sets up the problem, the related work, and the data model. **Part II** describes the ingestion pipeline (Task 1): system architecture and storage, data collection, contract–post linking, and text enrichment. **Part III** presents the analytical study (Task 2): descriptive analysis of contracts and discourse, signal–market correlation, cross-platform comparison, and the dashboard. **Part IV** reports the predictive study (Task 3), discusses why social signals fail to add value, states the limitations, and concludes.

CHAPTER 2

Background and Related Work

2.1 Prediction markets and informational efficiency

The idea that a market price aggregates dispersed private information into a single public statistic goes back to Hayek's argument that prices are society's mechanism for pooling knowledge no individual holds in full [**?**]. Modern prediction markets operationalise this idea directly: the Iowa Electronic Markets have forecast US election outcomes more accurately than polls across several cycles [2], and the broad survey of Wolfers and Zitzewitz [20] documents the same pattern across political, economic, and sports events. The theoretical appeal rests on the same logic as the efficient-market hypothesis [6]: if a price fails to reflect available information, an informed trader can profit by correcting it, and in doing so moves the price. Empirically, prediction-market prices have been found to be well-calibrated and hard to beat over long horizons [2, 20]. For our purposes this has a sharp methodological implication: *any* public signal that is genuinely predictive is also, by the same argument, likely to be *already incorporated* into the price. The bar for "incremental value" is therefore high by construction, and a negative result is the theoretically expected outcome unless the social signal carries information the market has not yet absorbed (for instance, an early or private signal).

2.2 Social media as a predictive signal

The hypothesis that aggregate online sentiment anticipates collective outcomes was popularised by Bollen et al. [4], who reported that Twitter mood dimensions improved short-horizon stock-index forecasts, and by Salathé and Khandelwal [16] in the public-health domain. Tetlock's study of media pessimism and the stock market [19] is more cautionary: he finds that media sentiment predicts *temporary* price pressure that subsequently reverts, consistent with sentiment proxying for noise rather than fundamentals. The broader literature is mixed, and many positive findings have proven fragile out of sample. PolySignal is designed precisely to avoid the methodological traps — look-ahead leakage, in-sample feature selection, and the conflation of correlation with incremental value — that produce fragile claims.

2.3 Sentiment versus stance

A recurring confusion in this literature is between *sentiment* (the emotional polarity of a text, independent of any target) and *stance* (the directional position of a text *with respect to a specific proposition*). "This is terrible news for Iran" is negative in sentiment but may *support* the YES outcome of a contract such as "Will Iran's regime fall?". Because the research question is about contract outcomes, we compute both: sentiment as a generic descriptive feature (Chapter 7) and outcome-aware stance as the directional signal that feeds prediction (Chapters 7 and 13).

2.4 Positioning

Relative to prior work, this project is distinguished less by a novel model than by *methodological discipline at Big Data scale*: a heterogeneous, million-row corpus; a transparent, deterministic linking and stance layer; a knowledge graph as a queryable projection; and an evaluation whose leakage controls are explicit and auditable. The headline result — that public social signals do not add incremental value here — is consistent with the efficiency argument above and is reported without embellishment.

CHAPTER 3

Data Model and Dataset Construction

The dataset has two interconnected components — **Polymarket contract data** and **social-media discourse data** — collected, linked, and organised into a single schema. This chapter defines that schema, maps it onto the fields required by the assignment, and describes how the 300-contract study set was selected. The mechanics of *how* the data is collected and processed are deferred to Part II; here we specify *what* the data is.

3.1 The unified schema

All normalised entities are stored as flat, typed tables (Apache Parquet; see Chapter 4 for the rationale). Tables 3.1 and 3.2 list the principal fields and show that every field required by the assignment is present, alongside the derived, analysis-ready columns we add.

Table 3.1. Contract and price-series schema. The left block lists the fields required by the assignment; the right block lists the value or derivation in our dataset.

Required field	Our column	Notes / derivation
markets.parquet — one row per contract		
`market_id`	`market_id`	Polymarket `conditionId`
`question`	`question`	full question text
`category`	`thematic_category`	{politics, finance, sports}; plus a finer `domain`
`outcomes`	`outcomes`	binary YES/NO
`resolution_outcome`	`resolution`	YES/NO declared at settlement
`resolution_date`	`resolved_at`	UTC timestamp
`creation_date`	`created_at`	UTC timestamp
`close_date`	`close_time`	UTC timestamp
`url`	`source_url`	canonical Polymarket URL
—	`volume`, `liquidity`	trading volume and book liquidity (derived)
price_points.parquet — one row per (contract, outcome, timestamp)		
`market_id`	`market_id`	foreign key to the contract
`outcome`	`outcome_id`	CLOB token id (YES/NO)
`timestamp`	`timestamp`	UTC; full-lifetime, ∼hourly to daily
`price`	`price`	implied probability in $[0, 1]$
`volume`	`volume`	cumulative volume where available

Table 3.2. Social-post schema (`social_posts.parquet` and the derived `post_enrichment.parquet`). "available" marks fields present where the source exposes them.

Required field	Our column	Notes
`post_id`	`post_id`	stable per-platform id
`platform`	`platform`	`telegram` / `reddit` / `twitter`
`url`	`url`	canonical post URL
`market_id`	via `linked_posts`	semantic link (Chapter 6)
`search_query`	`query`, `query_type`	query + generator type
`collected_at`	`collected_at`	retrieval timestamp
`category`	inherited from contract	through the link
`author_id/name`	`author_id`, `author_name`	available
`author_followers`	—	not exposed by broadcast channels
`published_at`	`created_at`	UTC, normalised
`text`	`text`, `title`	full text
`hashtags`	`hashtags`	regex-extracted (enrichment)
`mentioned_entities`	`mentioned_entities`	NER (enrichment)
`language`	`language`	detected (enrichment)
`like/reply/repost/view`	`engagement_score`, `upvotes`, `comments_count`, ...	available per platform
`comments`	`comments_count`, `post_comments`	counts always captured; a *sample* of reply text + author + timestamp is collected by the comment collector (Section 5.2)

3.2 The 300-contract study set

The assignment requires *at least 100 resolved contracts per thematic domain*. We assembled a **domain-balanced set of 300 resolved binary contracts — exactly 100 politics, 100 finance, and 100 sports** — under the three-way `thematic_category` used throughout the report. Internally, Polymarket exposes a finer six-value domain taxonomy; Table 3.3 shows how it maps to the assignment's three buckets and the resulting composition of the study set.

Table 3.3. Mapping from Polymarket's fine domain taxonomy to the three thematic categories, with the composition of the 300-contract study set.

Thematic category	Fine domains	Contracts
politics	politics, geopolitics	100
finance	crypto (56), economy/macro (44)	100
sports	sports (98), culture/entertainment (2)	100
Total		**300**

Each contract was required to be (i) *resolved*, so that the ground-truth outcome is known; (ii) *binary* YES/NO; and (iii) backed by a *verified full-lifetime price timeline* from creation to settlement. Contracts without a usable price history were rejected and replaced rather than padded. The resulting price store contains **1,353,114 price points** across all 300 contracts, with a median of **4,899** points per contract — dense enough to study intraday dynamics.

Figure 3.1 summarises the study set. The resolution distribution is strongly imbalanced: **243 contracts (81%) resolved No** and only **57 (19%) resolved Yes**. This base rate is itself informative — most "Will X happen by date D?" contracts describe salient but ultimately low-probability events — and it conditions the entire predictive analysis in Part IV.

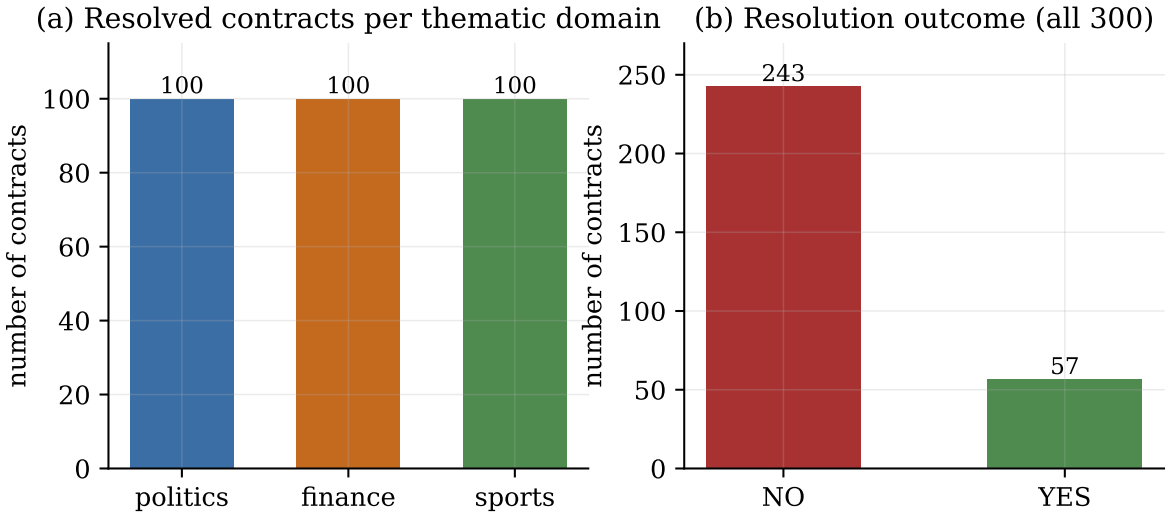


Figure 3.1. The 300-contract study set. (a) Exactly 100 resolved contracts per thematic domain. (b) The resolution outcome is heavily skewed towards NO (243 vs. 57), an imbalance that any predictive model must respect.

3.3 Dataset at a glance

Table 3.4 consolidates the headline figures of the constructed dataset; each is derived in the chapters that follow. Together they make the scale concrete: two heterogeneous sources, more than 2.7 million primary records before enrichment, and a fully linked and labelled analytical layer on top.

Table 3.4. The constructed dataset at a glance. Every figure is regenerated from the persisted pipeline artifacts (`scripts/_paper_figs.py`).

Artifact	**Count**	**Notes**
Resolved binary contracts	300	100 / 100 / 100 by domain
Price points	1,353,114	full lifetime, 300/300 contracts
Social posts (deduplicated)	1,368,041	Telegram 1,348,965 + Reddit 16,254 + X 2,822
Distinct channels / subreddits	68	—
Enriched posts	1,368,041	sentiment + NER + hashtags + language (all posts)
Scored contract–post pairs	4,112,547	deterministic relevance
Accepted evidence links	37,562	over 147 contracts
Outcome-aware stance labels	35,679	over 140 contracts
LDA topics	10	over linked discourse
Knowledge-graph relationships	> 2.7M	Neo4j projection

CHAPTER 4

System Architecture and Storage

This chapter describes the architecture of the ingestion pipeline (Task 1) and motivates the storage solution, which the assignment singles out as a graded design choice.

4.1 Architecture overview

PolySignal is a sequence of loosely coupled modules, each with a typed input contract, a typed output contract, and no knowledge of how its neighbours are implemented. Data flows in one direction — from external sources to the dashboard — and is persisted at every stage, so any stage can be re-run, audited, or replaced in isolation. Figure 4.1 shows the flow.

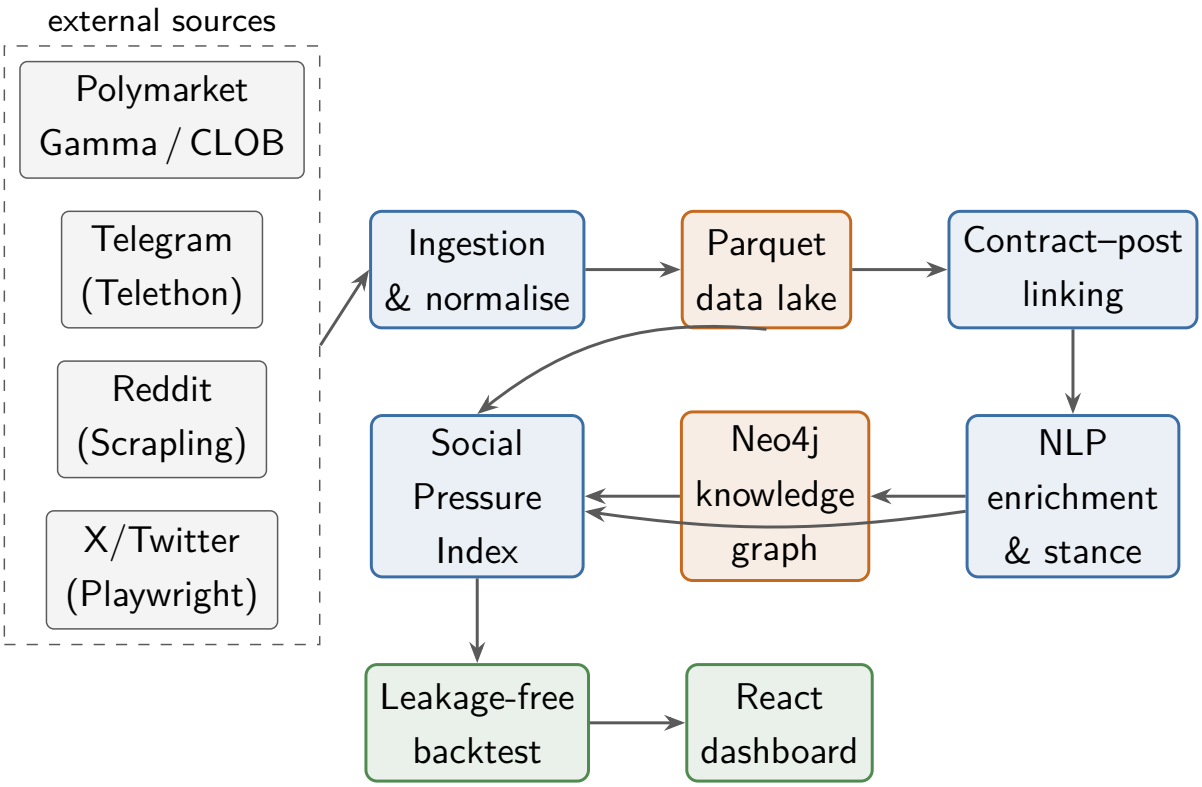


Figure 4.1. End-to-end architecture. Blue nodes are processing stages, orange nodes are stores, green nodes are user-facing outputs. Parquet is the analytical source of truth; Neo4j is a queryable projection of it.

4.2 Storage solution and its motivation

The dataset has three characteristics that drive the storage choice: it is *strongly temporal* (price series and longitudinal posts), it is *schema-heterogeneous* across platforms, and it is queried mostly by *wide columnar scans and aggregations* (group-by domain, by month, by market) rather than by point look-ups. We adopt a **two-store design**.

Apache Parquet as the analytical source of truth. All normalised and analytical tables are stored as Apache Parquet files [1]. Parquet is a good fit for this workload for several concrete reasons:

- **Columnar layout.** Analyses read a few columns over millions of rows (e.g. `price` over time, `sentiment_compound` by platform); a columnar format reads only those columns and skips the rest, with per-column compression that shrinks the ∼1.37M-post corpus from gigabytes of JSON to a few hundred megabytes.
- **Schema-on-write with evolution.** A single `SocialPostRecord` schema absorbs Reddit and Telegram, whose native shapes differ; new optional columns (the enrichment fields) are added as *derived tables* keyed by `post_id` rather than by rewriting the corpus, preserving the raw data.
- **Engine portability.** The same files are read by pandas/PyArrow on a laptop and by Apache Spark [21] at scale, with no migration. The "data zones" below are laid out so that a Hive-style partitioned write (`platform=/domain=/date=`) is an additive change.

The data lake is partitioned by *maturity* into zones, summarised in Table 4.1. This separation keeps immutable provenance (`raw/`) apart from validated records (`normalized/`) and from analysis-ready features (`analytical/`), which is what makes the pipeline auditable and re-runnable.

Table 4.1. Data zones, partitioned by maturity.

Zone	Purpose	Format
`raw/`	API payloads and social snapshots with retrieval lineage	JSON / JSONL
`normalized/`	validated, typed, deduplicated records	Parquet
`analytical/`	feature tables, evidence, timelines, backtests	Parquet / JSON
`cache/`	LLM responses and expensive intermediates, keyed deterministically	JSON / Parquet
`graph_exports/`	offline graph dumps for backup	CSV / JSONL

Neo4j as a relationship projection. Questions such as "which authors bridge the crypto and macro discussions?" or "what is the stance neighbourhood of this contract?" are graph traversals, awkward to express as columnar joins. We therefore project the Parquet tables into a **Neo4j** property graph [15], which is canonical for relationship analysis but *not* a replacement for time-series storage — price points deliberately stay in Parquet. The schema is detailed in Section 4.3.

> **Design choice.** Parquet for analytical, columnar, temporal scans; Neo4j for relationship traversal. Each store is used only for what it is best at, and Parquet remains the single source of truth.

4.3 Knowledge-graph schema

The Neo4j projection models the domain as a property graph (Table 4.2). Contracts, outcomes, posts, authors, platforms, and domains are nodes; linking and stance become typed, evidence-

bearing relationships. A `:LINKED_TO` edge is created only for *accepted* evidence (the policy of Chapter 6). Directional stance is modelled with a *single* `:SUPPORTS` edge that points from a post to the specific `:Outcome` node it argues for: a post backing No becomes a `:SUPPORTS` edge to that contract's No outcome, so the edge's *target* already encodes the direction and a separate `:OPPOSES` type is unnecessary. Only `supports_yes`/`supports_no` stances are projected; neutral, ambiguous, and insufficient-context rows are reported as skipped rather than projected. Price points stay in Parquet, since time-series analysis does not benefit from millions of graph nodes. Loading is idempotent (batched `MERGE` on `UNWIND` payloads), so the graph can be rebuilt from the Parquet tables at any time.

Table 4.2. Knowledge-graph contents derived from the current pipeline artifacts (the loader is idempotent over the Parquet tables, so each edge count equals its artifact count). Directional stance is a single `:SUPPORTS` edge to the supported outcome node, which removes the need for a redundant, always-empty `:OPPOSES` type.

Relationship	Count	Relationship	Count
`:BELONGS_TO` (contract→domain)	300	`:LINKED_TO` (post→contract)	37,562
`:HAS_OUTCOME`	600	`:SUPPORTS` (post→outcome)	6,569
`:POSTED_ON` (post→platform)	1,368,041	`:WROTE` (author→post)	1,367,960

CHAPTER 5

Data Ingestion

5.1 Polymarket contracts and price series

Contracts are collected from Polymarket's public **Gamma** API (market metadata) and **CLOB** API (price history) [14]. A binary filter keeps only YES/NO contracts; multi-outcome and team-vs-team markets are detected, logged with a reason, and skipped. The normaliser maps Polymarket identifiers to our schema (`conditionId`→`market_id`, per-outcome CLOB token ids →`outcome_id`) and writes the three contract tables of Table 3.1.

Full-lifetime price reconstruction. A practical obstacle is that the CLOB `/prices-history` endpoint returns only a narrow window for a settled market and rejects ranges longer than ∼two weeks. To obtain a *complete* probability trajectory — which Task 2 requires — the ingester walks each contract's lifetime in 14-day windows and merges them, upserting on `(market_id, outcome_id, timestamp)` so repeated fetches never duplicate samples. Contracts for which a continuous timeline cannot be reconstructed are reported and excluded.

5.2 Social-media collection

Social posts are collected longitudinally so that, for each contract, discussion spanning creation-to-resolution is available.

Telegram (primary). Public channels and groups are read through the official MTProto API using Telethon [9]. Collection pages backwards through each channel's history by message id, checkpointing after every batch so a killed run resumes exactly where it stopped; Telegram's rate limits (`FloodWait`) are honoured automatically. Telegram supplies the overwhelming majority of the corpus (Table 5.1).

Reddit (secondary). Reddit is collected via compliant public-web scraping with Scrapling [17] against `old.reddit.com`; the official OAuth API is not used. When Reddit returns `HTTP 403/429`, the collector opens a circuit breaker and respects a cooldown — it reports the block rather than evading it — so Reddit coverage is built up opportunistically over repeated, rate-limited passes.

X/Twitter (third platform, small but real). X is collected through **authenticated browser automation** (Playwright [11]): a logged-in session driven from market-derived search queries, with the same checkpointing and rate-limiting discipline as the other collectors. X's anti-automation defences (token challenges, login walls, and aggressive `HTTP 429` rate limiting) cap how much can be gathered responsibly, so X is a **small third platform — 2,822 posts** (Table 5.1) — rather than a major contributor. It is nonetheless genuinely present: X posts flow

through the same platform-agnostic schema and enter the cross-platform analysis (Chapter 11), where 114 contracts carry all three sources.

Comments and replies. For a sample of high-evidence posts the collector also retrieves a bounded sample of *replies/comments* — their text, author, timestamp, and per-comment engagement — satisfying the `comments` field of the schema.

Why no synthetic padding. Where a source is sparse, the corpus stays sparse. We never synthesise posts to inflate coverage; the consequences of this honesty are analysed in Chapter 15 rather than hidden.

Table 5.1. Composition of the analysed social corpus (deduplicated). All three platforms are present; Telegram dominates, while X/Twitter is a small third platform whose scale is capped by its anti-automation defences (Section 1.4).

Platform	Posts	Share
Telegram	1,348,965	98.6%
Reddit	16,254	1.2%
X/Twitter	2,822	0.2%
Total	**1,368,041**	100%

5.3 Market-aware query generation

Reddit retrieval and the targeted Telegram searches are driven by queries generated *from each contract question*. A naive keyword extractor would discard the load-bearing constraint in a question such as "Will China invade Taiwan *before GTA VI*?". The generator instead decomposes a question into an *event core*, its *constraints*, and *context anchors*, and emits a small, deduplicated family of query types — *core*, *entity*, *constraint*, *anchor* (including Roman↔Arabic numeral variants), *slug*, *domain*, *tag*, and *polymarket*. Each query carries trace metadata (`market_id`, `query_type`, `rank`) so that retrieval provenance is preserved on disk and can later inform — but never by itself decide — relevance.

Table 5.2. Query family generated for the contract *"Will China invade Taiwan before GTA VI?"*. The generator preserves the load-bearing *constraint* (the `before`-anchor) that a naive keyword extractor would discard, and emits numeral variants so that "GTA 6" and "GTA VI" both match.

Query type	Generated query
core	`china invade taiwan`
entity	`China`, `Taiwan`
constraint	`china taiwan before gta vi`
anchor	`GTA VI`, `GTA 6` (Roman ↔ Arabic variant)
slug	`will-china-invade-taiwan-before-gta-vi`
domain	`geopolitics`, `china taiwan war`

5.4 Normalisation, timestamps, and deduplication

Every ingested record is validated against a Pydantic model, its timestamp is normalised to timezone-aware UTC, and it is deduplicated. The pipeline preserves one row per `(post, query, candidate market, retrieval window)` tuple and flags later occurrences of the same `(platform, post_id)` as duplicates, so that a post surfaced by several queries is neither double-counted nor lost as additional retrieval evidence. After this step the corpus contains no duplicate posts.

CHAPTER 6

Contract–Post Linking

A retrieval query that mentions "Iran" returns many posts that have nothing to do with a *specific* contract about Iran. Linking is the step that decides, semantically, which posts are genuinely about which contract — and it is graded by the assignment as a key design choice.

6.1 A deterministic relevance scorer

The default linker is fully deterministic: no language model is required to reproduce any number in this report. For each candidate `(contract, post)` pair it computes a relevance score from interpretable signals:

- **entity and term overlap** between the contract question (and its slug/description) and the post text, weighted by salience;
- **constraint matching** — the temporal or contextual constraint of the question (a date, a threshold, an anchor event) must be compatible with the post, so that a post about "Bitcoin at $80k" is not linked to a "$150k" contract;
- **domain consistency**, penalising cross-domain false matches; and
- **temporal admissibility** — a post published after the contract closed can never be accepted as evidence.

Each pair receives a relevance label in {`relevant`, `weakly_relevant`, `ambiguous`} and a calibrated `confidence` in $[0, 1]$. Concretely, the base score is a weighted blend of four lexical-semantic overlaps,

$$\text{score} = 0.25\,J + 0.35\,T + 0.20\,E + 0.20\,Q \;+\; (\text{gated bonuses}) \;-\; (\text{penalties}), \tag{6.1}$$

where J is the Jaccard overlap of token sets, T the salience-weighted term overlap, E the named-entity overlap, and Q the overlap with the generating query; constraint matches add gated bonuses, while domain mismatch and temporal inadmissibility apply penalties. Labels follow fixed cut-offs on the resulting confidence (weakly-relevant at ≥ 0.60, relevant higher), and an optional, cached LLM verifier can be enabled for borderline rows — but it was *not* used for the results reported here: **all 4,112,547 candidate pairs were scored by the deterministic baseline**.

6.2 From candidates to accepted evidence

Scoring a pair is not the same as accepting it. The *acceptance policy* — applied identically by the graph loader and the evidence exporter — admits a pair as real evidence only if it is `relevant`, or `weakly_relevant` with `confidence` ≥ 0.60; *and* it resolves to an existing parent post and contract; *and* it is temporally admissible (pre-close, timestamp present); *and* it survives deduplication (per `(contract, post)` only the highest-confidence row is kept). Figure 6.1 shows

the resulting funnel from the raw corpus down to accepted evidence.

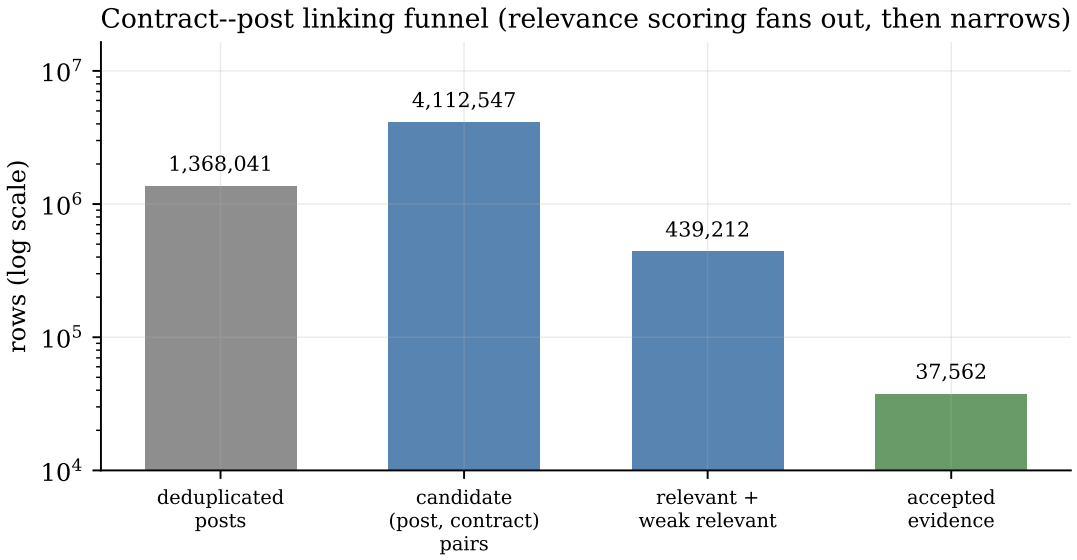


Figure 6.1. The linking funnel (log scale). Relevance scoring fans the corpus out into **4,112,547** candidate (post, contract) pairs, of which only **37,562** survive the acceptance policy as evidence. The largest losses are the unrelated majority and the weak links below the confidence floor.

Figure 6.2 explains why the funnel narrows so sharply: the bulk of scored pairs sit in the 0.40–0.50 confidence band, well below the 0.60 acceptance threshold. This conservative floor trades recall for precision — a deliberate choice, since downstream the cost of a spurious link (a post wrongly attributed to a contract) is higher than the cost of a missed one.

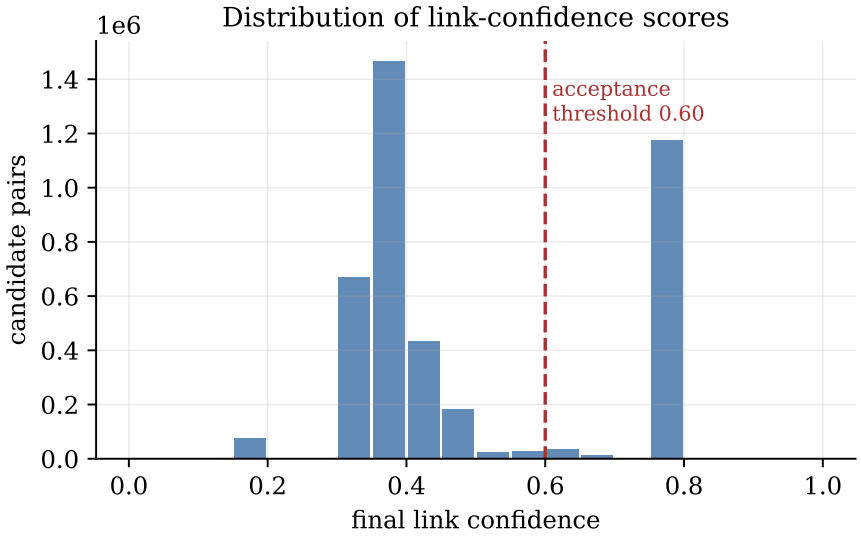


Figure 6.2. Distribution of final link-confidence scores over all 4,112,547 candidates. The dashed line marks the 0.60 acceptance threshold for weakly-relevant links; most candidates fall below it.

6.3 Coverage statistics

The assignment asks for coverage statistics — the fraction of contracts with at least k associated posts. Table 6.1 reports the relevance-label breakdown, and Figure 6.3 the depth of accepted evidence per contract. Accepted evidence reaches **147 of 300 contracts (49%)**; 35 contracts have at least 100 accepted links, while 153 have none. This uneven coverage is the central data-quality story of the project and is dissected in Chapters 11 and 15.

Table 6.1. Relevance labels over all 4,112,547 scored candidate pairs. Broad retrieval makes *unrelated* the majority label; the acceptance policy then keeps a small, high-precision evidence set.

Relevance label	Pairs	Share
relevant	26,337	0.6%
weakly_relevant	412,875	10.0%
ambiguous	150,352	3.7%
unrelated	3,522,983	85.7%
Total scored	**4,112,547**	100%
Accepted as evidence	**37,562**	0.9%

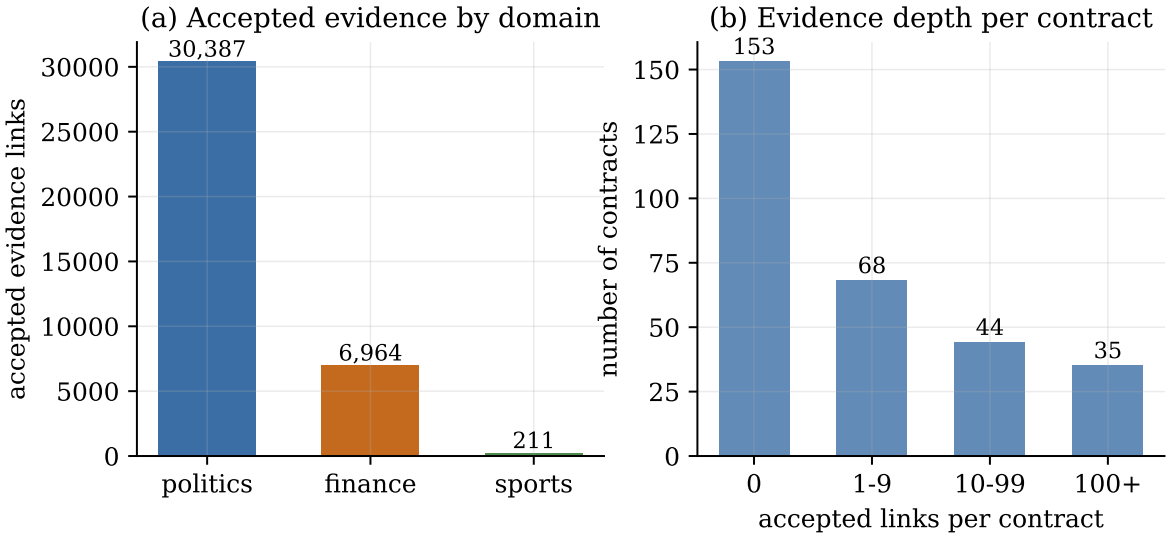


Figure 6.3. Evidence coverage. (a) Accepted evidence is dominated by the politics domain (30,387 links) and almost absent for sports (211). (b) Depth of accepted evidence per contract: 153 contracts have no accepted evidence, 35 have ≥ 100 links.

CHAPTER 7

Text Preprocessing and NLP Enrichment

Enrichment turns raw post text into the structured fields required by the assignment (hashtags, mentioned entities, language) and the analytical signals used later (sentiment, topics, stance). To respect the data-safety convention of never rewriting the ∼300 MB raw corpus, all enrichment is written to *derived tables* keyed by `post_id`, exactly like the linking and stance outputs. Every backend degrades gracefully so the pipeline always runs offline, and the chosen backend is recorded per row.

7.1 Normalisation, language, hashtags, entities

Text is lower-cased and stripped of control characters for matching while the original is retained for display; timestamps are normalised to UTC (Chapter 5). **Language** is detected with `langdetect` when available, otherwise a Cyrillic/stop-word heuristic; English dominates (1,118,413 enriched posts), followed by Russian (105,395), reflecting the geopolitics channels. **Hashtags and mentions** are extracted by deterministic regular expressions. **Named entities** are extracted with spaCy [7] when installed, otherwise a heuristic gazetteer-plus-capitalised-n-gram backend; the run reported here used the heuristic backend over all **1,368,041** enriched posts. Its limitations (channel handles and news boilerplate leaking in as pseudo-entities) are discussed in Chapter 9.

7.2 Sentiment analysis

Sentiment is computed with **VADER** [8], a rule-based model tuned for social text (emoji, capitalisation, slang) and a sensible default that needs no training data. Each post receives positive/neutral/negative proportions and a `compound` score in $[-1, 1]$, thresholded at ± 0.05 into a label. A transformer backend (`twitter-roberta`) is available as an opt-in upgrade. The distribution of sentiment is analysed in Chapter 9; here we note only that sentiment is a *generic* polarity signal, deliberately distinct from stance.

7.3 Topic modelling

Topics are extracted with **Latent Dirichlet Allocation** [3] over `CountVectorizer` features (scikit-learn [13]), run over the *linked* posts — the discourse that actually concerns the contracts. The vectoriser keeps the top **2,000** alphabetic terms (token pattern `[a-zA-Z]{2,}`, unigrams), drops terms appearing in more than 60% of documents or in fewer than three, and removes both the standard English stop-word list and a curated set of social/market boilerplate (e.g. “http”, “rt”, “amp”, ticker chatter). LDA is then fit with **ten topics**, the `"batch"` learning

method, **20** iterations, and a fixed `random_state=42`; each topic is summarised by its twelve highest-weight terms. We report ten topics as a deliberate, interpretable granularity rather than tuning topic count by a coherence criterion — the topics are used qualitatively, to characterise *what* the linked discourse is about, not as a predictive feature. The resulting topics and their interpretation are presented in Chapter 9.

7.4 Outcome-aware stance

Stance is the directional signal that feeds prediction, and it is computed *relative to each contract's outcomes*, not as generic polarity. For every accepted `(contract, post)` link the stance module interprets the post against the YES/NO proposition using outcome-specific cues, negation handling, and uncertainty markers, emitting a label in {`supports_yes`, `supports_no`, `neutral`, `ambiguous`, `insufficient_context`} with a confidence. The deterministic heuristic labelled **35,679** post–outcome rows across 140 contracts; Figure 7.1 shows the distribution.

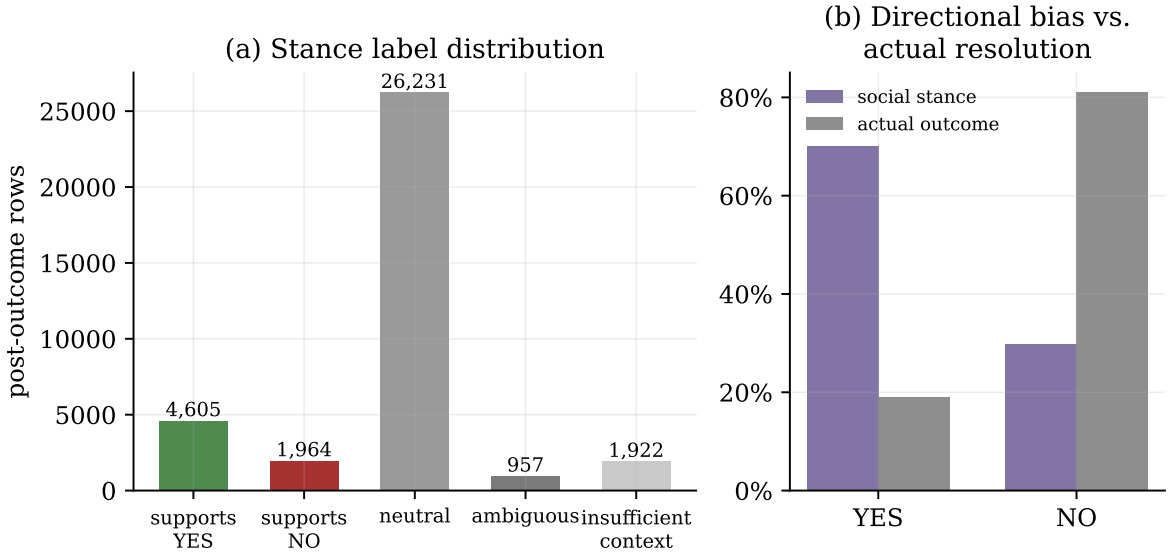


Figure 7.1. Outcome-aware stance. (a) Most rows are `neutral` (26,231); directional rows split 4,605 YES vs. 1,964 NO. (b) The social stance is **2.3:1 biased towards Yes**, whereas the contracts actually resolve **4.3:1 towards No** — the optimism/hype mismatch that recurs throughout the predictive analysis.

Sentiment vs. stance. Sentiment (VADER) measures the mood of the text; stance measures its directional position on a contract's outcome. Both are computed, but only stance feeds the forecast — and stance already reveals a systematic YES-bias that foreshadows the predictive null result.

CHAPTER 8

Descriptive Analysis of Polymarket Contracts

This chapter characterises the 300 contracts along the dimensions the assignment asks for — domain, duration, volume, and volatility — and then studies how market-implied probabilities evolve, and how informative they are of the final outcome at different distances from resolution.

8.1 Domain, duration, volume, and volatility

Table 8.1 summarises the three thematic domains, and reveals that they are qualitatively different objects despite the equal count.

Table 8.1. Descriptive profile of the contracts by thematic domain (medians; duration IQR in brackets).

Domain	Contracts	Duration (d)	Volume (USD)	YES-price vol.	% YES
politics	100	84 [28–181]	25.5 M	0.070	33%
finance	100	104 [31–134]	20.8 M	**0.114**	20%
sports	100	**258** [249–272]	**41.6 M**	**0.003**	4%
all	300	154	24.9 M	0.025	19%

Three patterns stand out (Figures 8.1 and 8.2):

- **Sports contracts are long-lived, high-volume long-shots.** They run for a whole season (median 258 days), attract the largest median volume, yet have the *lowest* price volatility (0.003) and resolve No 96% of the time — they are mostly "will team X win the league?" markets that sit near a stable, low probability for months and then resolve against the field.
- **Finance contracts are the most volatile** (0.114): crypto price-level bets ("Bitcoin above $\$X$") swing as the underlying asset moves.
- **Politics contracts are the shortest and most balanced** in outcome (33% Yes), consistent with sharp, event-driven questions.

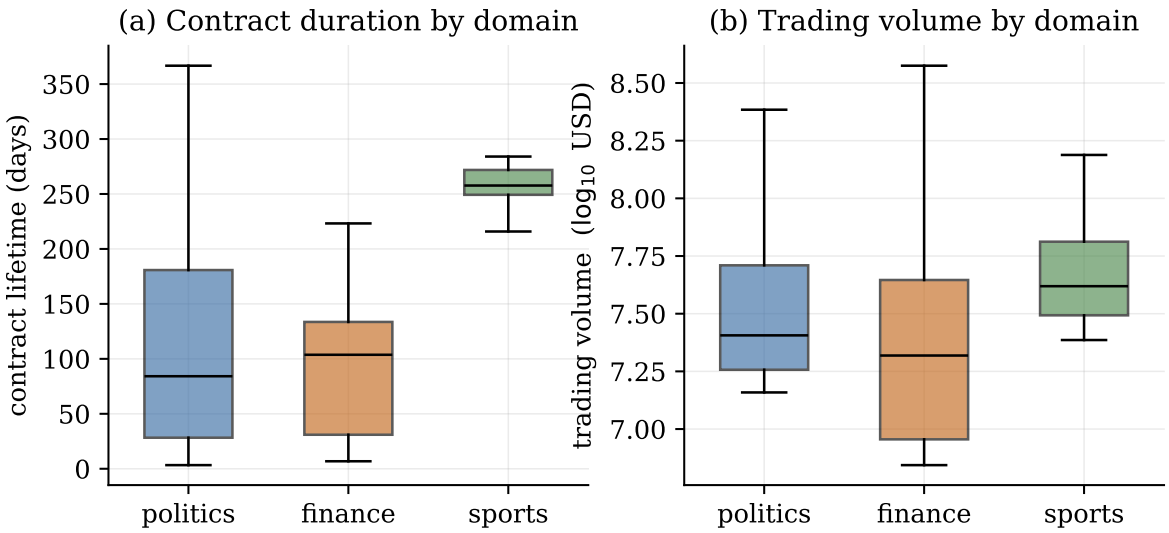


Figure 8.1. Contract lifetime (a) and trading volume (b) by domain. Sports contracts are markedly longer-lived; volumes are large across the board (note the $\log_{10}$ scale).

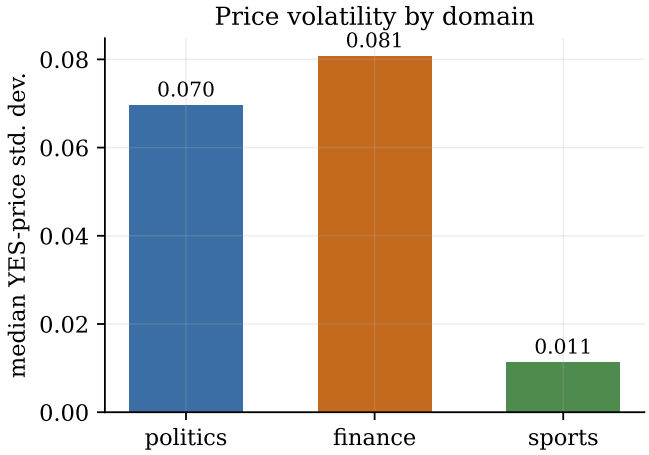


Figure 8.2. Median volatility of the YES price by domain. Finance is an order of magnitude more volatile than sports.

8.2 Are prices informative of the outcome?

The central descriptive question is whether the market price is informative of the final outcome, and how that informativeness depends on the distance to resolution. We take, for each contract, the latest YES price at least h days before close, treat it as a probabilistic forecast $\hat{p} = \Pr(\text{YES})$, and score it against the realised outcome with AUC and the Brier score $\frac{1}{N}\sum_i(\hat{p}_i - y_i)^2$ [5]. Figure 8.3 shows the result.

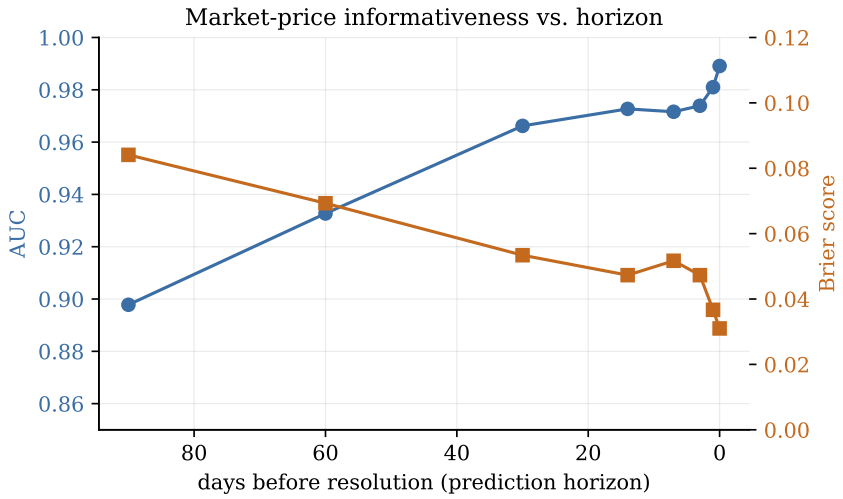


Figure 8.3. Market-price informativeness as a function of the prediction horizon (the x-axis runs from 90 days before resolution, left, to the close, right). AUC rises monotonically from 0.90 to 0.99 and the Brier score falls from 0.084 to 0.031 as resolution approaches.

The market is already a *strong* forecaster three months out (AUC 0.90) and becomes near-perfect at the close (AUC 0.99, Brier 0.031). This is the textbook behaviour of an efficient market resolving uncertainty over time [20], and it has a decisive consequence for the rest of the report: **the bar any social signal must clear — adding value on top of a price that is already this informative — is very high**. We return to this in Chapter 13.

> The market price alone separates YES from NO contracts with AUC 0.90 three months before resolution and AUC 0.99 at the close. Informativeness grows monotonically as the event nears.

CHAPTER 9

Social-Media Discourse Analysis

We now characterise the discourse: its volume and timing, its sentiment, the entities and topics that dominate it, and how these differ across platforms and domains (Task 2.2).

9.1 Volume, timing, and sources

The corpus spans a decade of channel history but is concentrated in recent years, with the analysis focused on the post-2024 window in which the studied contracts are active (Figure 9.1). It is overwhelmingly a **Telegram** corpus (98.6%); Reddit contributes 16,254 posts (1.2%) and X/Twitter a further 2,822 (0.2%), a small but real third platform whose scale is capped by its anti-automation defences (Section 1.4). Volume is concentrated in a handful of large channels (Figure 9.2): crypto exchanges and news aggregators (`CryptoComOfficial`, `cointelegraph`), macro/markets feeds (`zerohedge`, `FinancialJuice`, `WalterBloomberg`), and geopolitics/OSINT channels (`nexta_live`, `ClashReport`, `intelslava`, `rybar`). This channel mix is the single most important fact about the corpus: **it is a crypto/macro/geopolitics corpus, with essentially no sports coverage** — a mismatch we quantify in Chapter 11.

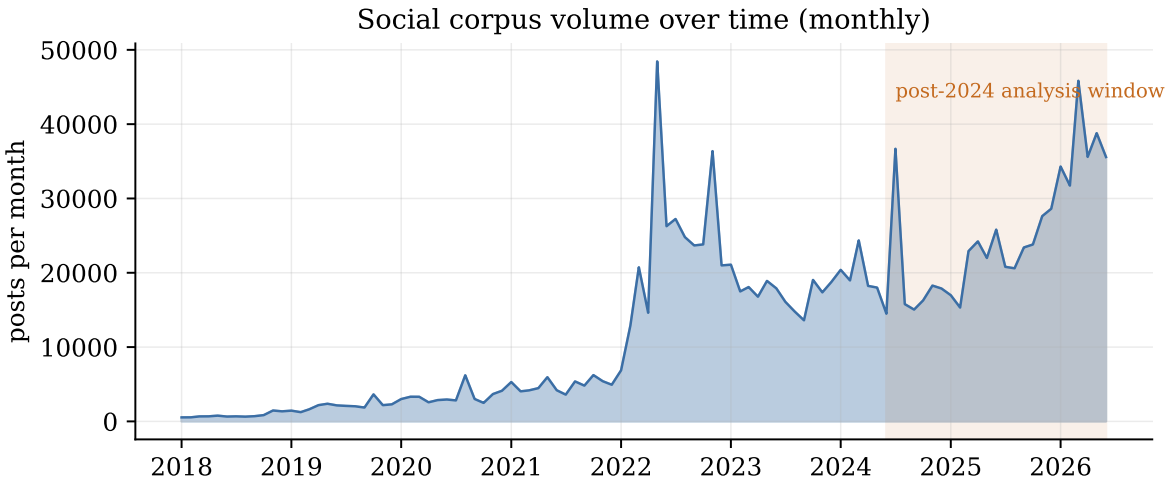


Figure 9.1. Monthly volume of the social corpus. Activity accelerates through 2024–2026; the shaded region marks the analysis window in which the studied contracts are live.

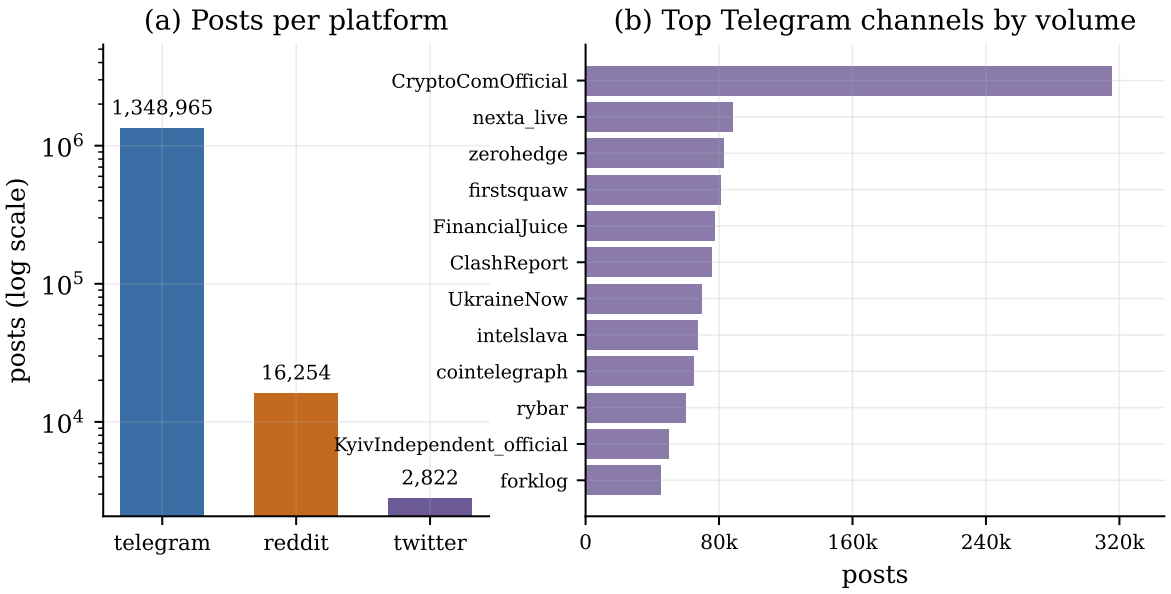


Figure 9.2. (a) Posts per platform (log scale): Telegram dominates. (b) The largest Telegram channels are crypto, macro, and geopolitics sources.

9.2 Sentiment

Sentiment, computed with VADER (Chapter 7), is close to balanced but slightly positive on aggregate (mean compound +0.006): **48% neutral, 28% positive, 24% negative** (Figure 9.3). The neutral mass is expected for a corpus heavy in news-wire and OSINT reporting ("BREAKING: ..."), whose register is factual. The platform breakdown is similar on Telegram and Reddit, so sentiment does not, by itself, distinguish the two sources.

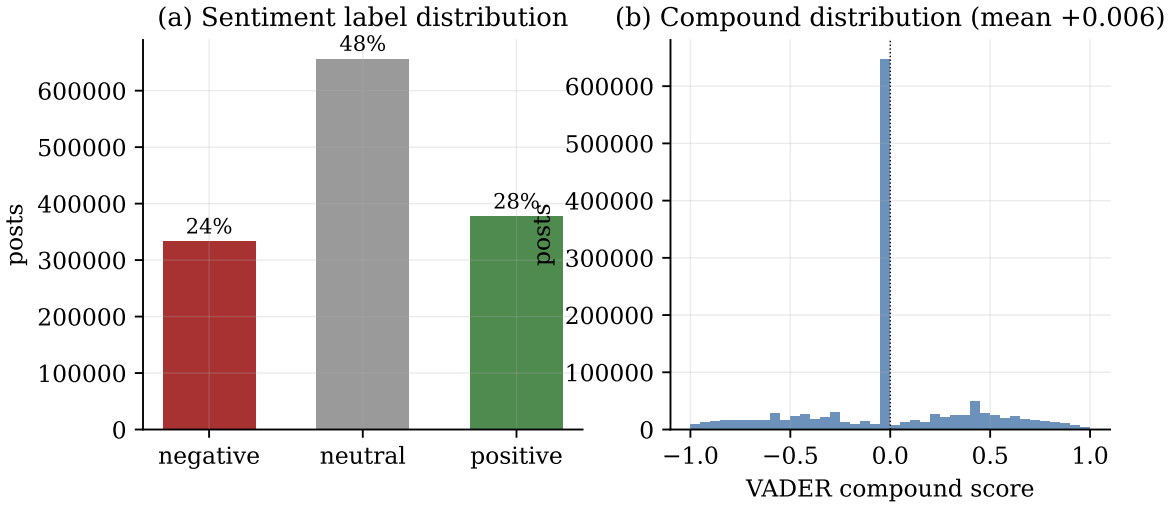


Figure 9.3. (a) Sentiment-label distribution over 1,368,041 enriched posts. (b) The VADER compound score is centred near zero with a slight positive tilt.

9.3 Named entities

The most frequent entities (Figure 9.4) trace the corpus's preoccupations: *Trump*, the *US*, *Ukraine*, *Iran*, *Russia*, *Israel*, *Gaza*, the *Fed*, *China*, and *Bitcoin*. The list confirms a politics, geopolitics, macro, and crypto corpus, and the near-total absence of sports entities. It also exposes an honest limitation of the heuristic NER backend: a few high-frequency "entities" are in fact channel handles or news boilerplate (e.g. broadcast-footer tokens, app names such as *WhatsApp* and *Viber*, and markers like *Breaking* or *Just In*). We filter the most obvious of these from Figure 9.4; a production system would use the spaCy backend (available but not used at scale here) and a stop-entity list. The artefact does not affect the predictive results, which depend on stance, not on raw entity counts.

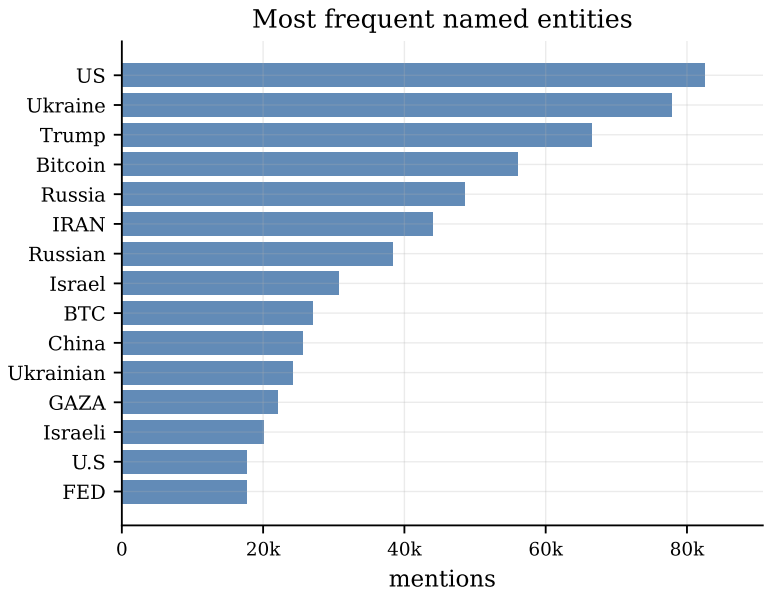


Figure 9.4. Most frequent named entities (obvious channel/app/boilerplate tokens removed). The corpus is dominated by geopolitics, macro, and crypto references.

9.4 Topics

Latent Dirichlet Allocation over the linked posts yields ten coherent topics (Figure 9.5 and table 9.1). They partition cleanly into *geopolitics* (Iran–Israel strikes, Iran nuclear talks, Trump and Ukraine, the Iranian regime), *macro* (Fed, rates, inflation), and *crypto* (Bitcoin, Ethereum, and trading groups). No topic is about sports. The topic structure thus mirrors the entity and channel analyses: three things the corpus talks about a great deal, and one of the three required domains it barely mentions.

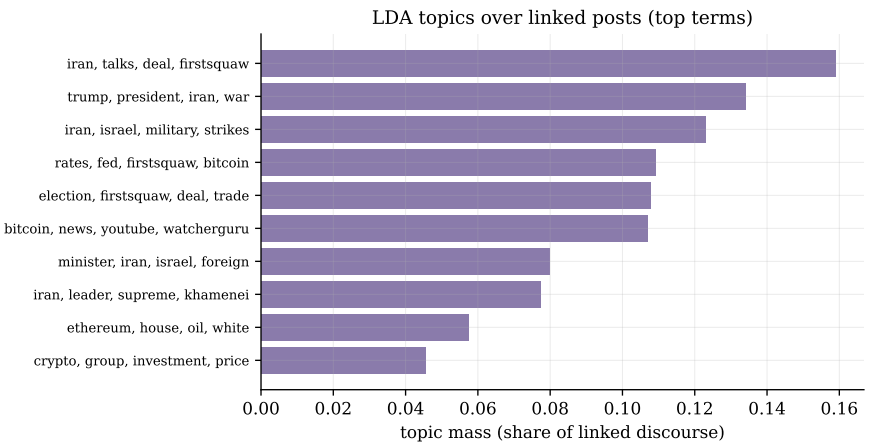


Figure 9.5. LDA topic mass over the linked discourse. Geopolitics, macro, and crypto topics dominate; sports is absent.

Table 9.1. The ten LDA topics over linked posts, by mass, with their leading terms.

Mass	Theme	Top terms
0.159	Iran nuclear talks	iran, talks, deal, trump, russia, nuclear, ukraine
0.134	Trump / war / Ukraine	trump, president, iran, war, ukraine, russia, ceasefire
0.123	Iran–Israel strikes	iran, israel, military, strikes, iranian, missile, regime
0.109	Fed / rates	rates, fed, inflation, policy, rate, hold, cut
0.108	Elections / trade	election, deal, trade, rates, fed, president, japan
0.107	Bitcoin / crypto news	bitcoin, news, btc, crypto, ethereum, saylor, bank
0.080	Diplomacy / ministers	minister, iran, israel, foreign, prime, rubio, security
0.077	Iran supreme leader	iran, leader, supreme, khamenei, iranian, killed
0.057	Ethereum / oil / WH	ethereum, house, oil, white, buterin, vitalik
0.046	Crypto trading groups	crypto, group, investment, price, sell, buy, discussion

CHAPTER 10

Signal–Market Correlation

This chapter studies the temporal relationship between the social signal and the market price (Task 2.3): does social activity precede, coincide with, or follow price moves, and is the direction of social stance aligned with the direction of price change?

10.1 Constructing the social signal

To compare social discourse against price on the same axis, we condense the accepted, directional evidence into a per-day social estimate of Pr(YES). Let w_t^+ and w_t^- be the day-t YES/NO evidence weights, where each stance contributes confidence $\times\,(1+\log(1+\text{engagement}))$, capped at 4.0 per post and at 3.0 per source per market-day so that no single author or channel dominates a market-day (source diversity), and the daily totals are EWMA-smoothed with a 3-day half-life. The social signal and its strength are then

$$s_t \;=\; \frac{w_t^+ + \frac{\kappa}{2}}{w_t^+ + w_t^- + \kappa}, \qquad \sigma_t \;=\; \frac{w_t^+ + w_t^-}{w_t^+ + w_t^- + \kappa} \;\in [0,1), \tag{10.1}$$

a Laplace-smoothed estimate anchored at 0.5 with pseudo-count $\kappa = 1.5$ (implemented as a per-side $\alpha = 0.75$): on thin-evidence days $\sigma_t \approx 0$ and $s_t \approx 0.5$, so the signal does not spike on noise. We treat the social line as meaningful only where σ_t is well above zero.

10.2 Do social spikes lead, coincide with, or follow price moves?

We answer this quantitatively. On days with meaningful evidence ($\sigma_t > 0.2$; 1,848 contract-days over 66 contracts), the pooled Pearson correlation between s_t and the market probability is only **0.05**, and the median per-contract correlation is 0.09 (Figure 10.1b) — weak and scattered around zero. To test *direction*, we cross-correlate daily *changes*, Δs_t against Δp_{t+k}, for lags $k \in [-7, 7]$ pooled over the well-covered contracts (Figure 10.1a): a positive lag k means the price moves *after* the social signal, i.e. social would be *leading*. The profile is essentially flat — every lag stays within $|r| < 0.04$ — and the only mild peak sits at lag 0 (coincident), with **no positive-lead peak**. There is therefore no horizon at which social changes systematically precede price changes: the little structure that exists is coincident at best, consistent with discourse *reacting* to, rather than anticipating, the events the market prices.

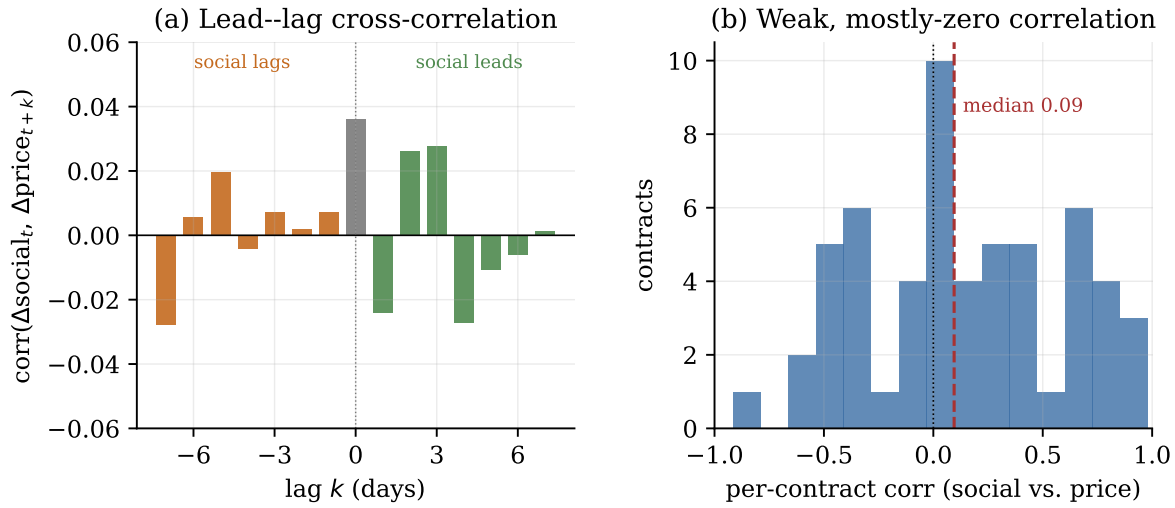


Figure 10.1. Social signal versus price, quantitatively. (a) Lead–lag cross-correlation of daily changes: a positive lag ($k > 0$) would mean social *leads* the price, yet every lag stays within $|r| < 0.04$ and the only mild peak is at lag 0 (coincident) — there is no positive-lead peak. (b) Distribution of per-contract social–price correlations: weak and centred near zero (median 0.09). Together they show the social signal neither leads nor meaningfully tracks the market price.

10.3 Is the direction of stance aligned with the outcome?

Aggregating stance into a per-contract direction (Equation (10.1) at the final pre-close cut), the directional agreement between the social signal and the realised outcome is only **0.21** — worse than a coin flip — and the correlation between the aggregate stance balance and the outcome is −0.015 (essentially zero). The reason is the optimism mismatch already visible in Figure 7.1: among the 72 contracts on which social pressure points YES, only 14 actually resolved YES (Figure 10.2). Social discourse is systematically biased towards the exciting YES outcome, while the contracts overwhelmingly resolve NO.

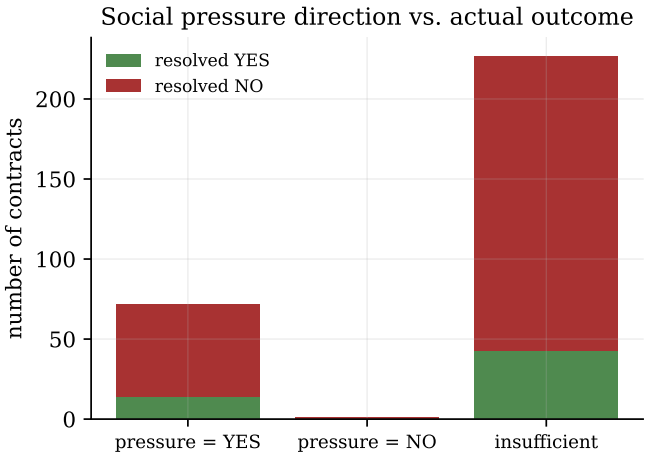


Figure 10.2. Social-pressure direction versus actual outcome. Where social pressure says YES (72 contracts), the market still resolves NO 58 times; the social direction is anti-correlated with reality.

Social activity neither leads nor strongly coincides with price (pooled correlation 0.05, flat lead–lag profile), and its *direction* is anti-aligned with outcomes (agreement 0.21). The discourse is reactive and optimism-biased.

CHAPTER 11

Cross-Platform Comparison

Task 2.4 asks three concrete questions: do the platforms carry *complementary* or *redundant* signals; are there contracts for which one platform is substantially more informative than the others; and what distinguishes those cases. This chapter answers each over all three collected platforms (Telegram, Reddit, and X/Twitter), with Telegram supplying the overwhelming majority of posts (Section 1.4). The cross-platform layer reads from `cross_platform_comparison.parquet` (one row per contract that carries social discourse) and its per-domain roll-up `cross_platform_summary.parquet`.

11.1 How many platforms cover each contract?

Of the 300 contracts, **232 carry at least one linked social post**, and **181 of those are multi-platform** — discussed on at least two of Telegram, Reddit, and X/Twitter within the analysis window, with **114 carrying all three**. Table 11.1 breaks the 232 down by domain, and Figure 11.1 visualises it.

Table 11.1. Platform coverage by domain over the 232 contracts that carry social discourse. *Single* = one platform present; *multi* = at least two platforms present; *compl.* = contracts the cross-platform analysis flags as genuinely *complementary* (a minority platform contributes non-trivially rather than echoing the majority).

Domain	With social	Single-platform	Multi-platform	Complementary
politics	85	21	64	1
finance	58	10	48	0
sports	89	20	69	**28**
total	**232**	51	181	29

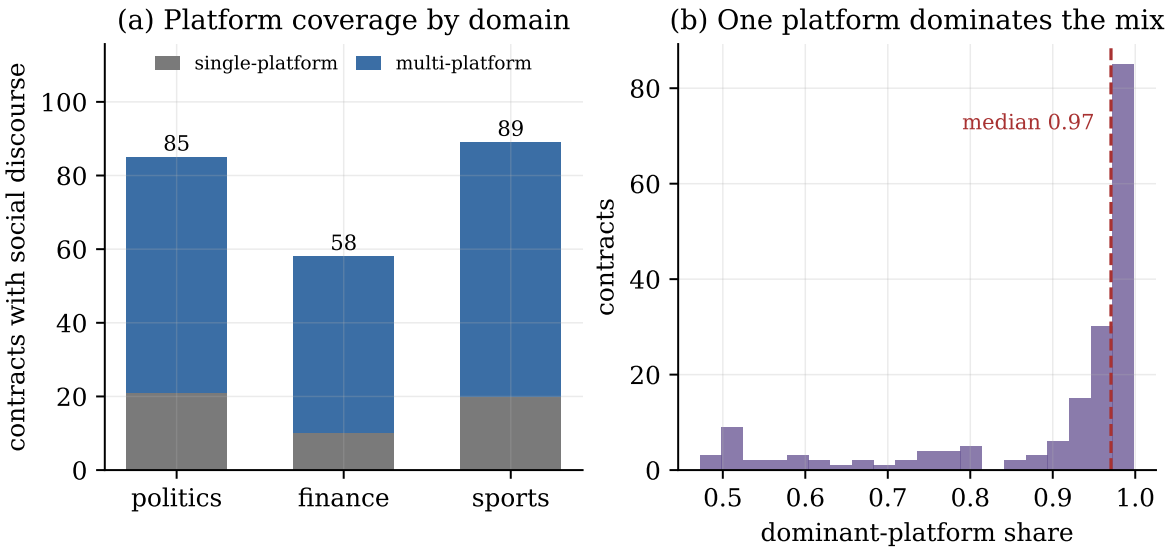


Figure 11.1. Cross-platform coverage. (a) Single- vs multi-platform contracts by domain: most covered contracts appear on both platforms. (b) Distribution of the *dominant-platform share* over multi-platform contracts — Telegram supplies the overwhelming majority of posts even when Reddit is also present, so the two signals are mostly **redundant rather than complementary**.

11.2 Redundant or complementary?

On the 181 multi-platform contracts the *dominant platform* (Telegram in almost every case) accounts for a median share of **∼0.97** of the posts: when several platforms speak about a contract, they say the same thing and Telegram says far more of it. By the standard meaning of the question, the signals are therefore **redundant, not complementary** — adding Reddit or X to Telegram rarely changes the per-contract social estimate. The lone exception is **sports**: 28 sports contracts are flagged complementary, precisely because Telegram barely covers sports (Section 11.5), so what little signal exists there comes disproportionately from Reddit. This is the one corner of the dataset where a second platform measurably adds coverage.

11.3 Where is one platform substantially more informative?

The per-contract view identifies the cases the assignment asks for. For *geopolitics and macro* contracts — US–Iran, Russia–Ukraine, the Fed — Telegram is overwhelmingly the more informative platform, contributing thousands of posts where Reddit contributes tens. For the handful of *sports* contracts that are covered at all, Reddit is relatively (though still thinly) the more informative source. No contract in the snapshot is substantially better covered by Reddit than Telegram in absolute terms; that asymmetry is a property of the collected channel mix, and the cross-platform table is the artefact that would surface the reverse cases under a broader collection.

11.4 A genuine three-platform comparison

Because every platform normalises to the same `SocialPostRecord` and the cross-platform analysis keys only on `platform`, all three sources enter Table 11.1 and fig. 11.1 uniformly: **114 of the 232 covered contracts carry posts from all three platforms**. X/Twitter's contribution is small in volume (Section 1.4), so Telegram still dominates the per-contract mix, but the comparison is genuinely three-platform rather than a two-platform result with a placeholder column. Broadening the X or Reddit pass would repopulate this chapter with no code change — re-running `polysignal analytics run` followed by the figure regeneration is all that is required.

11.5 The channel–market mismatch

The more consequential cross-cutting comparison is not between platforms but between the *topics* the channels discuss and the *domains* of the contracts. Table 11.2 lays the mismatch bare: politics contracts absorb 81% of all accepted evidence, finance most of the rest, and **sports almost none** (211 links across 100 contracts). The channels are crypto/macro/geopolitics sources; they simply do not talk about who will win the Premier League.

Table 11.2. Accepted evidence by domain, against the equal contract counts. Social evidence is concentrated where the channels' topics lie, not where the contracts are.

Domain	Contracts	Accepted links	Share of evidence
politics	100	30,387	80.9%
finance	100	6,964	18.5%
sports	100	211	0.6%
total	300	37,562	100%

The contracts for which social discourse is by far the most informative are therefore the *geopolitics* ones — US–Iran, Russia–Ukraine, Israel–Iran — which attract thousands of accepted links each, while crypto price-level and Fed-rate contracts attract hundreds and sports contracts attract a handful. This is the key qualitative finding of the cross-cutting analysis, and it directly explains the predictive results of Part IV: the social signal is dense and usable for one slice of the market and absent for another.

CHAPTER 12

Interactive Dashboard

All results are exposed through a custom interactive **React** [10] single-page application (designed and built with Claude's design tooling, *not* the suggested Streamlit [18] default, to obtain a more polished and customisable interface). It is served on a single local origin by a small, zero-dependency Python API (`scripts/serve_app.py`, launched by `run_app.cmd`) that materialises the exact persisted artefacts behind this report into the `window.PS_DATA` contract: the app needs *no internet, no CORS, and no build step*, never recomputes features, retrains models, or calls an external service, so a live demo is deterministic and fast. The application is the deliverable shown live during the oral discussion.

12.1 Views

The app is organised as two linked views plus a persistent customisation panel (Table 12.1). It opens on the *Market Universe*; selecting any contract drills into the *Market Detail*.

Table 12.1. The React dashboard: views, the panels they contain, and the Task-2 requirement each satisfies.

View / panel	What it shows
Market Universe	browse all 300 contracts as a card grid; filter by *domain* (macro, politics, crypto, AI, sports, culture) and *status* (active, closing, resolved); each card sparklines price vs. social signal
Market Detail	opens on the per-contract *dual-line chart*: market Pr(YES) vs. the smoothed social signal (Equation (10.1)), with event markers
Social panel	post volume, sentiment split, and the stance distribution for the contract
Scorecard / Divergence	headline outcome, resolution date, and where social pressure *diverges* from the price
Cross-platform	per-platform post mix for the contract (Chapter 11)
Evidence inspector	individual linked posts with stance tag, sentiment, entities, engagement, and a "why linked" rationale
Narrative graph	the contract's neighbourhood in the Neo4j knowledge graph
Tweaks panel	live customization: accent theme, layout density, chart style (pressure/area), and signal thresholds

12.2 Design rationale

The dashboard is designed directly against the five graded dashboard axes. *Priority*: it opens on the Universe and one click reaches the dual-line price-vs-social chart — the project's central comparison — so the main idea is never more than a click away. *Accuracy*: every panel reads the exact persisted artefact behind the corresponding number in this report, and thin-evidence contracts are flagged rather than smoothed (the social line is drawn only where σ_t

is meaningful, per Equation (10.1)). *Conciseness*: each panel answers one question and the card grid compresses 300 contracts into a scannable overview. *Usability*: domain/status filters, sparkline cards, and a single-contract drill-down make navigation immediate. *Customization*: the Tweaks panel lets the user re-theme, re-densify, switch chart styles, and re-threshold the signal live. The explicit browsing requirements of Task 2 — browse by domain, resolution date and outcome; visualise price alongside social activity; explore sentiment and engagement distributions; and inspect individual linked posts with their metadata and extracted features — are each served by one of the panels above.

Reconciling the dashboard counts with the report. Two deliberate display choices make some headline counts differ between the live app and the figures in this report, and we state them explicitly so the two artefacts are seen to agree. First, the app renders the **295** contracts that pass its display guards — a handful of finance contracts whose only candidate links are off-subject, or whose price series is degenerate, are held out of the *demo* so no misleading card is ever shown — while the full **300**-contract dataset and *every* analytic in this report use the complete set. Second, the app's "with-evidence" badge uses a deliberately stricter *on-subject, directional* bar (**66** contracts) than the linking-layer coverage quoted in Chapter 6 (**134** contracts with ≥ 1 accepted link), so that a social card is drawn only where the evidence is genuinely informative. The two views are consistent: they report coverage at different points of the same acceptance funnel (Figure 6.1), conservatively by design.

CHAPTER 13

Outcome Prediction from Social Features

This chapter undertakes the optional Task 3: it engineers social features, trains forecasting models under a leakage-controlled, temporally honest protocol, and asks directly whether social discourse improves on the market price. The answer is a clear *no*, and the value of the chapter is in establishing that answer rigorously, with an ablation that isolates exactly the incremental contribution of the social signal.

13.1 Feature engineering: the Task-3 feature matrix

Task 3 starts from a single feature matrix with one row per resolved contract (`task3_feature_matrix.parquet`, 300 rows), assembled through the same leakage-safe evaluation join used by the backtest. It realises all four feature families suggested by the assignment, each computed strictly from *pre-close, accepted* evidence:

Volume.
Total accepted posts, posting growth rate (late-vs-early log ratio over the contract lifetime), unique authors and channels, and active days. The comments count is an honest zero, as per-comment text is collected only as a bounded sample.

Engagement.
Aggregate, mean, max and log-total engagement, plus a high-vs-low-engagement ratio relative to the corpus median.

Sentiment.
Mean and variance of VADER compound sentiment, the positive / negative / neutral fractions, and an early-vs-late sentiment trend.

Linguistic.
A per-contract text document over the accepted post text, turned into a TF–IDF representation reduced to 16 latent dimensions by truncated SVD. Crucially this vectoriser is re-fit *inside each training fold*, so the linguistic representation never leaks across the temporal split.

On top of these we carry the interpretable per-market *Social Pressure Index* (SPI) features (Table 13.1), which condense the stance-labelled evidence into a compact, engagement-weighted directional signal, and the market arm contributes the latest valid pre-close YES price as its single predictor. Table 13.1 lists the SPI subset; the full matrix spans all four categories above.

Table 13.1. The eight social features (the SPI whitelist) and the assignment category each realises. Every feature is computed only from pre-close, accepted evidence.

Feature	Category	Definition
`stance_balance`	stance	net directional lean $(n_{\text{Y}} - n_{\text{N}})/(n_{\text{Y}} + n_{\text{N}})$
`pressure_strength`	stance	magnitude of the net (engagement-weighted) directional pressure
`temporal_decay_pressure`	temporal	net pressure weighted towards evidence near the close (recency)
`pressure_acceleration`	temporal	change in pressure between the early and late windows
`evidence_quality_score`	quality	mean link/stance confidence of the accepted evidence
`source_diversity_score`	volume	channel/author entropy (anti-spam source spread)
`accepted_evidence_ratio`	volume	accepted links / scored candidates for the contract
`engagement_log_total`	engagement	$\log(1 + \text{total engagement})$ over accepted posts

13.2 Models and evaluation protocol

We evaluate three nested feature sets in increasing order of the information they use — **market-only** (the pre-close price), **social-only** (the full volume + engagement + sentiment + linguistic + SPI matrix), and **market + social** (both) — against a Laplace-smoothed base-rate baseline. On every feature set we train **four model families**: a regularised logistic regression, a Platt-calibrated linear SVM, Gaussian naive Bayes, and a gradient-boosting ensemble (all class-balanced, seed 42). Holding the model fixed and varying the feature set turns this 4×3 grid into a full *ablation* that isolates the incremental contribution of the social signal across model families.

Because contracts are intrinsically ordered in time, we score them under an **expanding-window temporal cross-validation** that respects chronology and forbids leakage: contracts are sorted by close date and partitioned into five contiguous blocks; fold i trains on blocks $0..i$ and is scored on block $i{+}1$, so every prediction is issued by a model trained only on *earlier*-resolving contracts. This yields 250 strictly out-of-fold predictions (5 folds $\times 50$). The leakage controls are explicit and auditable: the feature table is rejected if it contains any target/outcome/resolution column; labels are joined only after the social features pass validation; the linguistic TF–IDF/SVD is re-fit per training fold; the market probability is strictly pre-close; and settled post-resolution prices are never used.

The assignment's required metrics are **accuracy**, **macro-F1**, and **AUC-ROC**; we add the Brier score [5] and Expected Calibration Error (ECE). Macro-F1 — the unweighted mean of the two per-class F1 scores — is the headline metric here, because under the 57:243 YES:NO class imbalance a model that always predicts NO already reaches 0.81 accuracy:

$$\text{macro-F1} = \tfrac{1}{2}\left(\text{F1}_{\text{YES}} + \text{F1}_{\text{NO}}\right), \qquad \text{Brier} = \tfrac{1}{N}\sum_{i}(\hat{p}_i - y_i)^2, \tag{13.1}$$

$$\text{ECE} = \sum_{b} \tfrac{n_b}{N}\,|\text{acc}_b - \text{conf}_b|. \tag{13.2}$$

For full reproducibility, Table 13.2 pins the exact configuration of every model family.

All estimators are class-balanced and share a fixed seed (`random_state` = 42); the linear SVM is wrapped in a sigmoid (Platt) calibrator so that it, too, emits probabilities for the Brier/ECE/AUC computations. Two estimators (Gaussian naive Bayes and gradient boosting) use the scikit-learn defaults deliberately: with only 250 out-of-fold points, tuning their many knobs would invite overfitting to the evaluation, so we fix sensible defaults and let the *feature set*, not hyperparameter search, carry the comparison.

Table 13.2. Exact model configuration (scikit-learn). All families use `class_weight="balanced"` and `random_state=42`. "default" means the scikit-learn default is used unchanged, by design.

Family	Configuration
Logistic regression	`C=0.5`, `solver="liblinear"` (L2 penalty), `max_iter=1000`
Linear SVM	`LinearSVC(C=0.5, max_iter=5000)` wrapped in `CalibratedClassifierCV(method="sigmoid", cv=3)`
Gaussian naive Bayes	defaults (no smoothing override)
Gradient boosting	defaults (100 stages, `max_depth=3`, `learning_rate=0.1`)
Linguistic features	`TfidfVectorizer` ($\leq$2000 features, unigrams) $\rightarrow$ `TruncatedSVD(16)`, re-fit *per training fold*
Cross-validation	expanding window, 5 contiguous folds by close date

13.3 Results: temporal-CV ablation

Table 13.3 and Figure 13.1 report the primary result. Across all four model families the **market arm is a strong forecaster** (macro-F1 0.90–0.93, AUC 0.93–0.95), while the **social-only arm collapses to near chance** (macro-F1 0.47–0.51, AUC 0.50–0.56, barely above the 0.446/0.510 base-rate baseline). Combining the two never beats the market arm: the best combined configuration (linear SVM, macro-F1 0.933) merely *ties* the best market-only configuration (macro-F1 0.933), and for naive Bayes the social features actively *hurt* (macro-F1 0.918$\rightarrow$0.655). The result is robust to the choice of model family.

Table 13.3. Task-3 outcome prediction over 300 resolved contracts under expanding-window temporal cross-validation (250 out-of-fold predictions). Macro-F1 and AUC-ROC for each model family $\times$ feature set; higher is better. Base-rate baseline: macro-F1 0.446, AUC 0.510. The best macro-F1 is bold.

	Macro-F1↑			AUC-ROC↑		
Model	mkt	soc	m+s	mkt	soc	m+s
logistic regression	0.911	0.500	0.904	0.929	0.515	0.912
linear SVM	**0.933**	0.467	0.933	0.928	0.499	0.914
naive Bayes	0.918	0.485	0.655	0.926	0.534	0.735
gradient boosting	0.901	0.510	0.897	0.954	0.562	0.945

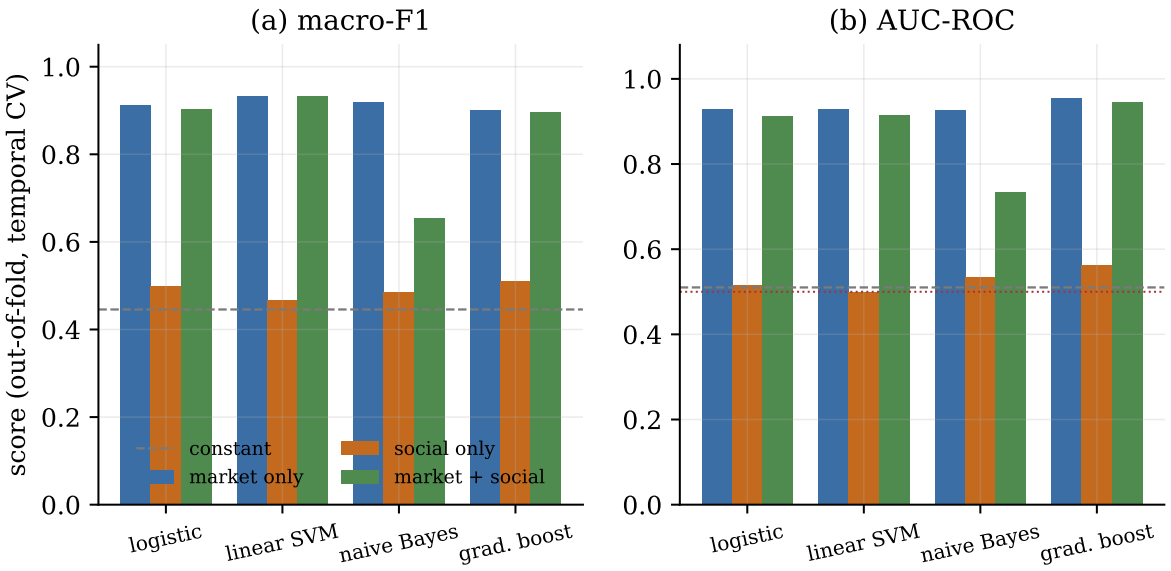


Figure 13.1. Task-3 temporal-CV ablation. (a) macro-F1 and (b) AUC-ROC for each model family, grouped by feature set (market only / social only / market + social). The market arm (blue) dominates across every family; the social arm (orange) sits at the constant baseline (grey dashed) and the chance line (red dotted); adding social to market (green) yields no consistent gain.

13.4 Complementary view: final pre-close backtest

As a calibration-focused cross-check we also keep the Branch-09 single-cut backtest — one final pre-close prediction per contract, scored under 5-fold out-of-fold cross-validation (Table 13.4, Figure 13.2). It tells the same story from the probabilistic angle: market-only attains Brier 0.031 / AUC 0.989, while market+social is slightly *worse* and social-only is below chance (AUC 0.404).

Table 13.4. Final pre-close backtest (5-fold out-of-fold; lower Brier/log loss, higher accuracy/AUC are better). The best probabilistic score in each column is bold.

Model	Brier↓	Log loss↓	Accuracy↑	AUC↑	ECE↓
base rate	0.154	0.486	0.810	0.487	**0.003**
market-only	**0.031**	**0.110**	**0.963**	**0.989**	0.020
social-only	0.160	0.504	0.810	0.404	0.060
market + social	0.040	0.187	0.963	0.975	0.091

Lift of market+social over market-only: $\Delta Brier = -0.009$, $\Delta LogLoss = -0.077$ *(both negative = worse).*

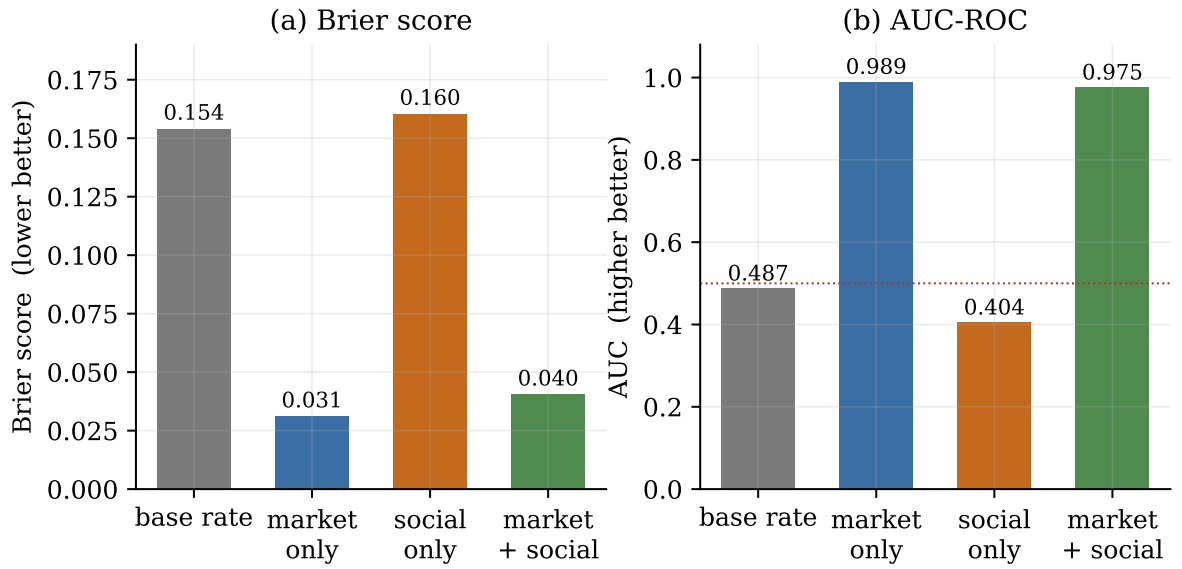


Figure 13.2. Final pre-close model comparison. (a) Brier score and (b) AUC. The market-only baseline (blue) is the best model; adding social features (green) degrades it, and social-only (orange) is no better than the chance line.

Calibration. Figure 13.3 shows the reliability of the two price-based models. Market-only is well calibrated (ECE 0.020): its predicted probabilities match observed outcome frequencies closely, in the sense of [12]. Market+social is visibly *less* calibrated (ECE 0.091) — the noisy social features push some predictions away from the well-calibrated price.

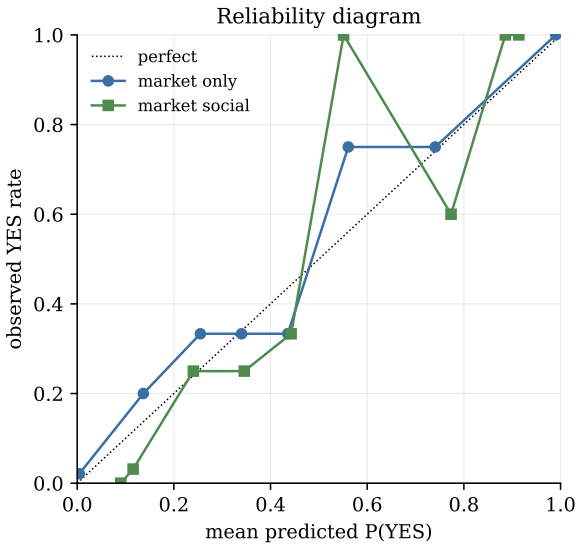


Figure 13.3. Reliability diagram. The market-only model (blue) hugs the diagonal; adding social features (green) worsens calibration.

Why the combined model does not improve. The fitted coefficients (Figure 13.4) tell the story compactly: the market-probability coefficient (1.68) dwarfs every social coefficient (all $|\beta| < 0.10$). The model learns to trust the price and to all but ignore the social features — and the small weight it does put on them adds variance, not signal.

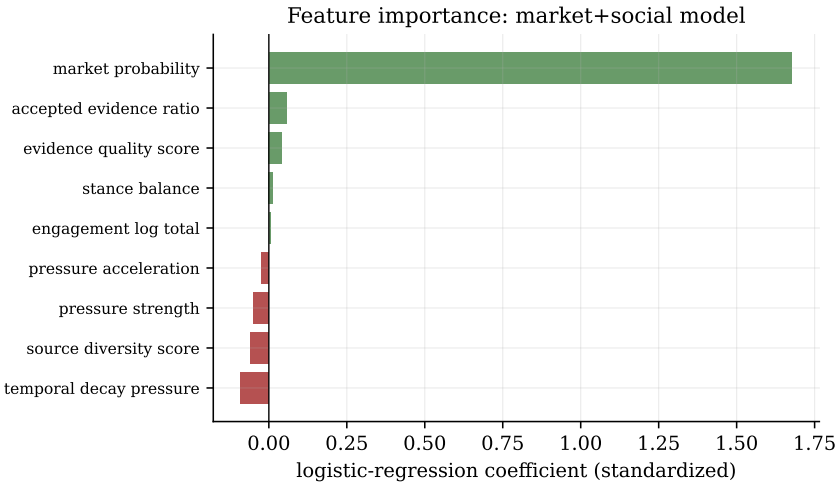


Figure 13.4. Standardised logistic-regression coefficients of the market+social model. The market probability dominates; the social features carry negligible, mixed-sign weight.

13.5 Does more evidence help? A coverage ablation

A natural objection is that the social signal fails only because coverage is thin. We test this by restricting the social-only model to the 73 contracts with *sufficient* accepted evidence. It does not rescue the signal: even there the model predicts No for every contract, so its nominal AUC of 0.56 reflects ranking noise on an all-No classifier rather than usable signal; on the 227 insufficient-evidence contracts the AUC is 0.35. The aggregate stance–outcome correlation is −0.015 and the engagement-weighted net-pressure–outcome correlation is −0.013 — both indistinguishable from zero. **Within the range of coverage we have, more evidence does not convert into predictive value**; the problem is the nature of the signal, not merely its quantity (we revisit the quantity question, honestly, in Chapter 15).

Headline result. Under expanding-window temporal cross-validation, all four model families place the market arm at macro-F1 0.90–0.93 / AUC 0.93–0.95, while the social-only arm sits at near-chance macro-F1 0.47–0.51 (AUC ≈ 0.5) and combining the two yields no consistent gain. The final pre-close backtest agrees: market-only scores Brier 0.031 / AUC 0.989, adding social makes it *worse*, and social-only is below chance (AUC 0.404). Social discourse adds *no* incremental predictive value over the Polymarket price in this dataset.

CHAPTER 14

Discussion

The predictive result is negative and robust. This chapter explains *why*, and argues that the finding is informative rather than disappointing.

14.1 Three structural causes

1. The market is already efficient with respect to public information. By the efficiency argument of Chapter 2, any genuinely predictive public signal tends to be arbitraged into the price. Section 8.2 quantified just how strong that price already is — AUC 0.90 a full quarter before resolution. A public, lagging social signal has essentially no room left to add value; the only edge would come from *private* or *early* information, which broadcast Telegram channels and public Reddit threads do not carry.

2. Social discourse is optimism-biased. Stance leans YES 2.3:1 while contracts resolve NO 4.3:1 (Figure 7.1), and on 58 of the 72 contracts where social pressure says YES the market resolves NO (Figure 10.2). Discourse concentrates on the exciting possibility that an event *will* happen; markets, pricing base rates, know that most such events do not. A signal that is systematically wrong in direction cannot help a calibrated forecaster.

3. The discourse is reactive, not leading. The lead–lag analysis (Chapter 10) found no horizon at which social changes precede price changes. Social media here narrates events the market has already priced; many candidate posts were rejected by the temporal-admissibility filter for falling *after* a contract had already closed. A coincident-or-lagging signal cannot improve a forecast made before the event.

14.2 A note on coverage

A fourth factor — the channel/market mismatch (Chapter 11) — limits how much of the market the social layer even *covers*: politics absorbs 81% of evidence, sports almost none. Crucially, though, the coverage ablation (Section 13.3) shows that the failure is not merely one of quantity: even where evidence is dense, its direction does not track outcomes. Broader coverage would let us test more contracts, but on present evidence there is no reason to expect it to overturn the direction of the result.

14.3 Why a null result matters

This is not a failed experiment; it is a clean, reproducible piece of evidence in a literature crowded with fragile positive claims. It (i) empirically corroborates the informational efficiency

of a major prediction market against a large, realistic social corpus; (ii) demonstrates a methodology — leakage controls, ablation, calibration — that distinguishes genuine incremental value from spurious correlation; and (iii) maps precisely *where* a social edge would have to come from (private/early signals, information-asymmetric events) for future work to find it. "PolySignal does not beat Polymarket" is, in this framing, exactly the finding.

CHAPTER 15

Limitations and Future Work

15.1 Limitations

We state the limitations plainly, since several of them bound the strength of the conclusion.

Platform coverage.
All three platforms are present in the analysed corpus, but it is **Telegram-dominated** (98.6%), with a secondary Reddit contribution (1.2%) and a small X/Twitter sample (0.2%) whose scale is capped by X's anti-automation defences (Section 1.4). The 68 Telegram channels are a purposive, not a random, sample, with a clear crypto/macro/geopolitics tilt. The predictive conclusions therefore hold for a Telegram-dominated corpus skewed towards crypto-adjacent discourse; whether a *volume-balanced* three-platform corpus would change them is the central open question, and the architecture is built to answer it without code changes once such a corpus is available.

Domain coverage.
Sports contracts have almost no social evidence; the predictive conclusion is therefore strongest for politics and finance and weakest for sports.

Comments and authors.
Only a bounded *sample* of per-comment text is collected, so comment-derived features are thin; broadcast Telegram channels expose neither follower counts nor author profile URLs, so those required fields are partially empty for that platform.

Deterministic baselines.
Linking and stance use deterministic heuristics; the optional cached LLM verifiers were not run at scale, which leaves 73% of stance rows `neutral` and the heuristic NER noisy.

One prediction per contract.
Although we now score under expanding-window temporal cross-validation across contracts, each contract still contributes a single pre-close prediction; intra-contract time-sliced 7/3/1-day forecasts are deferred until the price/feature history is broad enough to support them cleanly.

Binary contracts only.
Multi-outcome markets are out of scope.

15.2 Future work

The limitations map directly to next steps, ordered by expected payoff:

1. **Close the coverage gap.** Broaden the X/Twitter and Reddit passes (e.g. via a paid research tier) and add sports-specific channels, so that all three platforms and all three domains carry *comparable* evidence — turning the cross-platform layer of Chapter 11 into a volume-balanced three-platform complementarity test (Task 2.4).

2. **Stronger stance and linking.** Run the cached LLM stance/relevance judges at scale and adopt the spaCy NER backend, reducing the neutral mass and the entity noise.
3. **Time-sliced and early-signal evaluation.** Score 7/3/1-day pre-close windows and explicitly search for contracts where social *leads* the price — the only regime in which an edge is theoretically possible.

CHAPTER 16

Conclusions

We set out to determine whether social-media discourse adds predictive value over Polymarket's own prices. To answer it we built an end-to-end Big Data pipeline that collects 300 resolved binary contracts and their full-lifetime price series (1,353,114 points), collects a longitudinal multi-platform corpus (1,368,041 posts across Telegram, Reddit, and a small X/Twitter sample, Section 1.4), links posts to contracts with a deterministic semantic scorer, enriches the text with sentiment, entities, topics, and outcome-aware stance, projects everything into a Neo4j knowledge graph, and condenses the evidence into a per-market Social Pressure Index that feeds a leakage-controlled backtest — all exposed through an interactive dashboard. The system satisfies each requirement of Tasks 1, 2, and the optional Task 3.

The descriptive study confirmed that Polymarket prices are already highly informative of outcomes and become more so as resolution nears (AUC 0.90 $\rightarrow$ 0.99). The predictive study returned a clear, well-controlled **null result** that is robust across four model families under expanding-window temporal cross-validation: the market arm reaches macro-F1 0.90–0.93 and AUC up to 0.95, the social-only arm never escapes near-chance (macro-F1 $\approx$ 0.5), and combining the two adds no consistent lift; the final pre-close backtest concurs (market-only Brier 0.031 / AUC 0.989, market+social *worse*, social-only below chance). We traced this to the market's informational efficiency, a systematic optimism bias in social stance, the reactive (non-leading) nature of the discourse, and a structural mismatch between the available channels and the contract sample.

The contribution of this work is therefore twofold: a reproducible, leakage-aware methodology for studying social signals against prediction markets at scale, and the honest empirical finding that public social discourse does not, in this setting, beat the market price. In a field prone to over-claiming, demonstrating *where* and *why* a signal fails — and doing so reproducibly — is a result worth reporting.

CHAPTER A

Reproducibility and How to Run

The project is self-contained and portable: it targets **Python 3.11+**, declares every dependency in `pyproject.toml`, and the dashboard runs fully offline from the persisted pipeline artifacts — *no API keys and no internet access are required* to reproduce the figures, tables, and the interactive app in this report. Secrets live only in an untracked local `.env`; the shared `.env.example` is a blank template. The steps below reproduce the deliverable on a clean machine.

1. Create an isolated environment and install. The core install is enough to launch the dashboard and run the figure/test suite; the bracketed extras add the optional collectors and developer tooling.

```
python -m venv .venv
# Windows:        .venv\Scripts\activate
# macOS / Linux:  source .venv/bin/activate
pip install -e .                          # core: dashboard + analysis
pip install -e ".[dev,nlp,telegram,scraping]"   # everything (optional)
```

2. Launch the interactive dashboard. A single command builds the served payload from the artifacts in `data/` and opens the browser on a local, zero-dependency origin (CORS-free, no external calls).

```
python scripts/serve_app.py        # cross-platform
run_app.cmd                        # Windows one-click equivalent
# -> http://127.0.0.1:8000/
```

3. Regenerate every figure and table. All quantitative results are derived from the persisted artifacts by a single released step, so nothing in this report is hand-transcribed.

```
python scripts/_paper_stats.py     # refresh the stats JSONs from data/
python scripts/_paper_figs.py      # writes pdf_latex/figures/*.pdf
pytest                             # run the test suite (757 passing)
polysignal --help                  # explore the pipeline CLI
```

4. (Optional) rebuild the report PDF. The `pdf_latex/` folder compiles with any standard TeX toolchain — `pdflatex + bibtex`, `latexmk`, `tectonic`, or Overleaf — using only standard packages.

```
cd pdf_latex && latexmk -pdf main.tex     # or: tectonic main.tex
```

What the release bundle contains. The publishable archive ships the source (`src/`, `scripts/`, `webapp/`, `tests/`), the report (`pdf_latex/` with `main.pdf`), the documentation (`docs/`), and the *minimal* set of `data/` artifacts the dashboard reads (timelines, social pressure, stance labels, accepted evidence, market metadata, post enrichment, and the analytics/backtest

summaries — about 60 MB). The multi-gigabyte raw corpus, caches, virtual environment, and portable tool downloads are deliberately excluded; they are regenerable and carry no analytical value beyond what the shipped artifacts already summarise.

CHAPTER B

AI-Assisted Development: Tools and Repositories

In the interest of transparency, this appendix documents the AI-assisted development workflow behind PolySignal. The *scientific* results are independent of it — every number in this report is regenerated deterministically from the persisted artifacts (Appendix A), and no AI agent ever produced a result: the agents wrote *code*, the code produced the artifacts, and the artifacts produced every figure and table. What the AI workflow bought was engineering throughput *with* discipline.

B.1 AI coding assistants

Three command-line coding agents were used in deliberately complementary roles, so that no single model's blind spots went unchecked:

Claude Code (Anthropic).
The primary agent for implementation, refactoring, and this report. It carried the bulk of the pipeline — ingestion, contract–post linking, outcome-aware stance, the Social Pressure Index, and the leakage-controlled backtest — drove the test-first workflow, and authored and revised the LaTeX of this document.

Codex (OpenAI).
A second pair of eyes for focused code generation and cross-checking: alternative implementations of tricky functions, and independent review of edge cases and failure modes.

Gemini CLI (Google).
Used for large-context sweeps across the codebase and for corroborating design decisions, providing a third independent perspective on architecture and data-handling choices.

Running three independent agents — rather than trusting one — turned model disagreement into a useful signal: where they agreed we proceeded with confidence; where they diverged, the divergence flagged exactly the spots that warranted a human decision.

B.2 Supporting repositories and tooling

ECC — a comprehensive "everything Claude Code" framework.
ECC is an open repository — the winning entry of a 2026 hackathon — that packages a complete Claude Code setup: agent definitions, multi-agent orchestration rules, reusable skills and commands, and a full set of engineering conventions (coding style and immutability, a mandatory test-coverage bar, security checks against hard-coded secrets and unvalidated input, and a code-review gate before any change is accepted). We adopted ECC to drive and standardise the agentic workflow; it is what kept three different agents producing consistent,

reviewable code rather than three divergent styles, and why the repository ships with **757 passing tests** and a uniform module layout.

Graphify.

A repository-graph tool (configured through `.graphifyignore`, output under `graphify-out/`) that builds a navigable map of the codebase — its modules, symbols, and their relationships. It gave the agents structural context for navigating a large repository, so edits respected the existing architecture (Chapter 4) instead of reinventing it.

B.3 How the tools were leveraged

The workflow composed these pieces into a single loop: plan a change with one agent, implement it test-first, cross-review it with a second agent, and keep every change behind the ECC review-and-coverage gate, using the Graphify map to land edits in the right place. The result is a large, well-tested, well-documented codebase and a fully reproducible report, produced quickly but without surrendering the scientific honesty that is the whole point of a null-result study. The AI tooling accelerated the *engineering*; the *evidence* remains entirely the data's.

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